

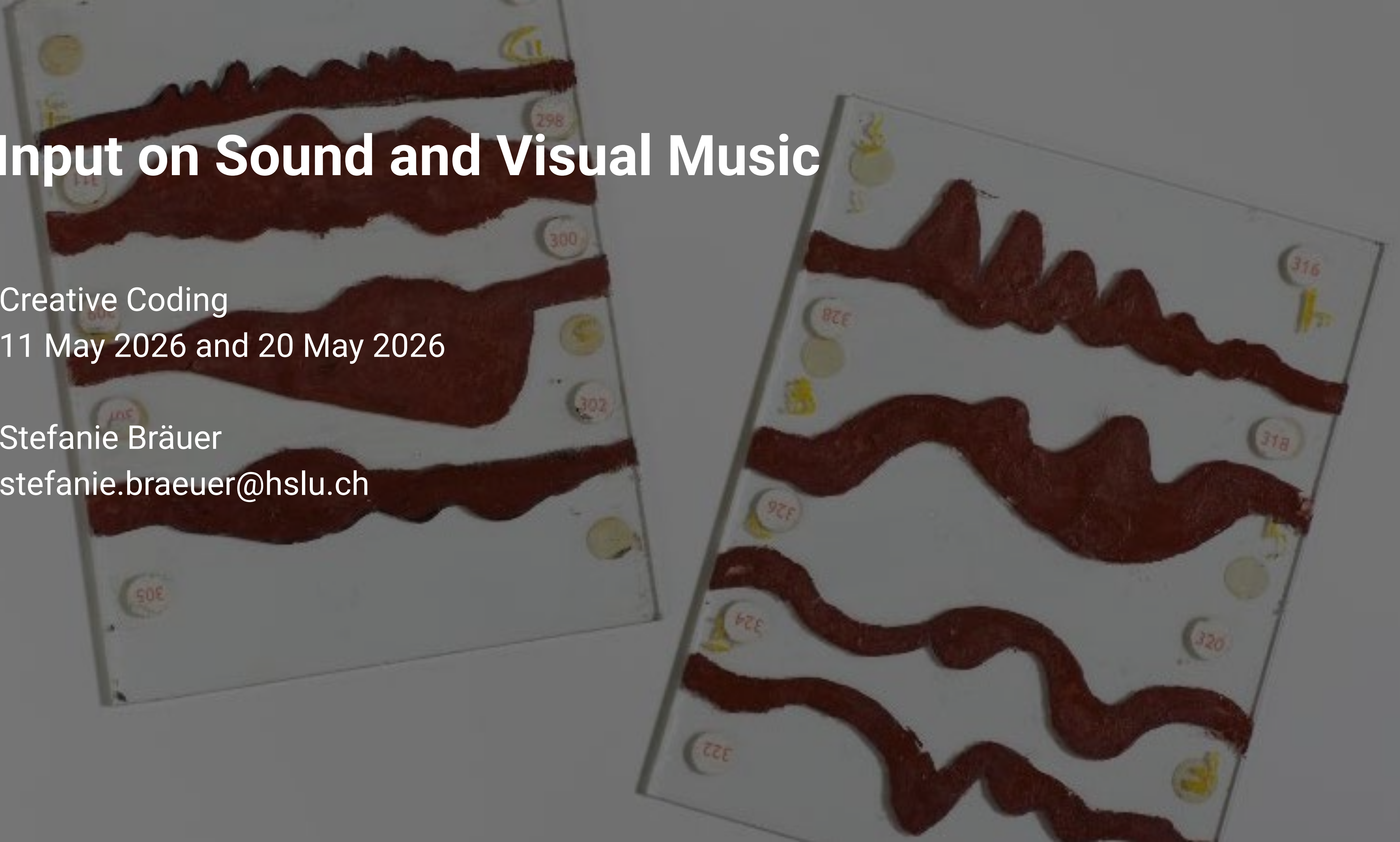
Input on Sound and Visual Music

Creative Coding

11 May 2026 and 20 May 2026

Stefanie Bräuer

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11 May 2026:

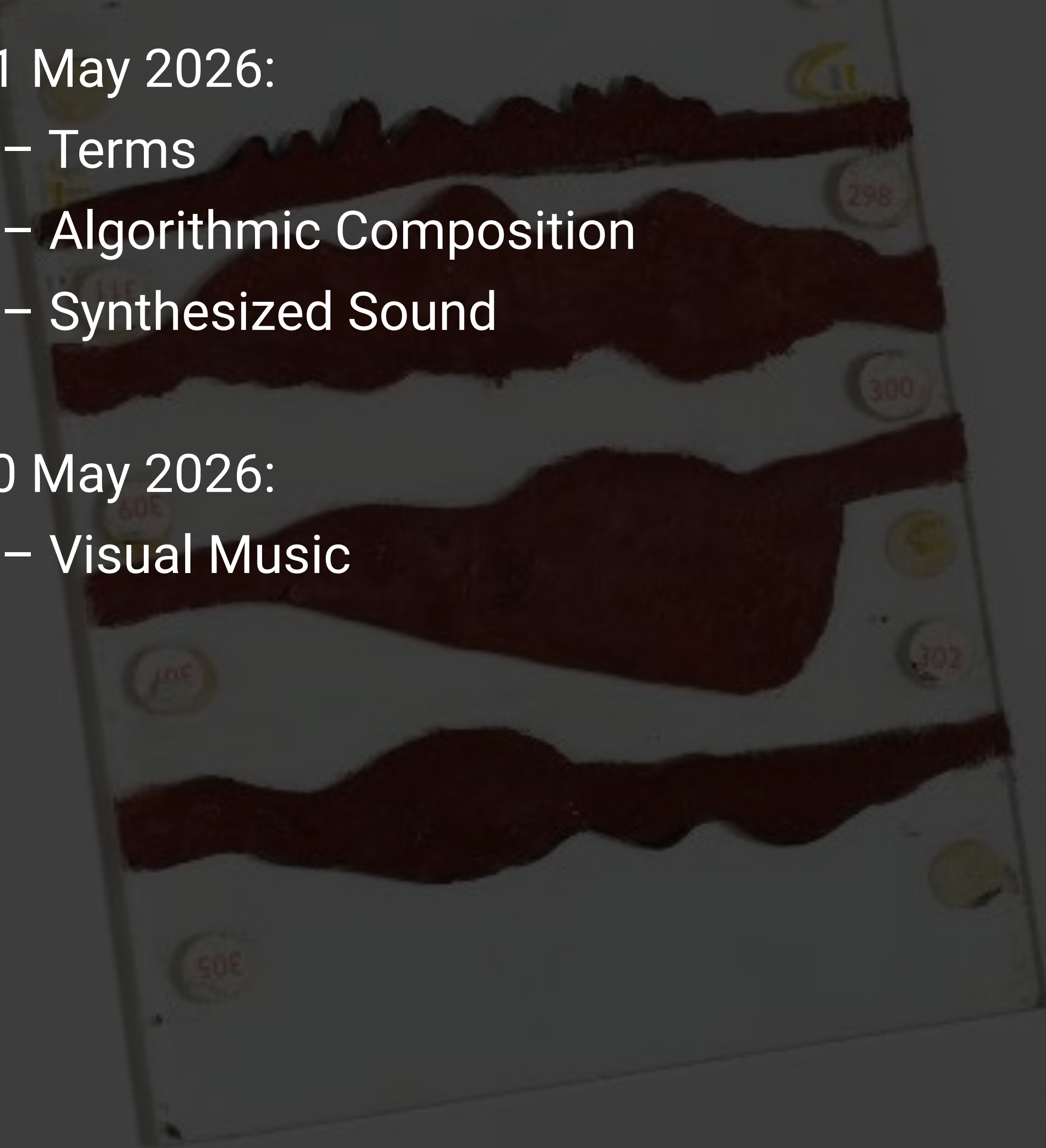
1 – Terms

2 – Algorithmic Composition

3 – Synthesized Sound

20 May 2026:

4 – Visual Music



1 – Terms

Differences between:

- sound (Schall vs. Klang), noise (Geräusch), tone (Ton), music
- layers of physics (acoustics), physiology, psychology, culture

„**Klang** als physikalische Eigenschaft des **Schalls** definiert sich durch ein jeweils charakteristisches Muster von Obertönen [...]. Dieses korrespondiert wiederum mit der psychophysischen Wahrnehmung, welche die Basis für die kulturelle Formung von Klängen [...] bildet. Erst auf dieser Ebene findet sich schließlich **Musik** als, sagen wir, abgrenzbares System kommunikativen Handelns oder kulturelles Dispositiv.“

Großmann, Rolf: „Die Materialität des Klangs und die Medienpraxis der Musikkultur. Ein verspäteter Gegenstand der Musikwissenschaft?“, in: Volmar, Axel und Jens Schröter (Hrsg.): Auditive Medienkulturen. Techniken des Hörens und Praktiken der Klanggestaltung, Bielefeld: transcript 2013, S. 61–77, hier S. 64., hier S. 63.

Sound als „das Klingende in seiner allgemeinsten Form.“

Ebd. S. 64.

„Traditionelle Notation als Medium der Gestaltung verweist nicht auf den Klang, sondern primär auf den **Ton** [...] Periodizität, seine Frequenz oder musikalisch gesagt, seine Tonhöhe.“

Ebd. S. 65.

1 – Terms

Soundscape/Sonosphere

„The **soundscape** is any acoustic field of study. We may speak of a musical composition as a soundscape, or a radio program as a soundscape or an acoustic environment as a soundscape.“

Schafer, R. Murray. 1994. The Soundscape. In The Soundscape: Our Sonic Environment and the Tuning of the World. Rochester, Vermont: Destiny Books. pp. 3–12.

„Unlike R. Murray Schafer’s notion of **soundscape**, [Pauline] Oliveros’s **sonosphere** embraces a full sweep and barrage of energies, including the magnetic, electrical, electromagnetic, geomagnetic, and quantum, as well as the acoustical.“

Kahn, Douglas: „Pauline Oliveros. Sonosphere“, in: Earth Sound Earth Signal. Energies and Earth Magnitude in the Arts, Berkeley: University of California Press 2013, S. 174–186, hier S. 174.

1 – Terms



Pauline Oliveros – Hörbeispiel aus *Deep Listening*, 1988

1 – Terms



Christina Kubisch: *Electrical Walks*, 2012 (mit Anfängen in den 1970er Jahren)

Der
allezeit fertige
Polonoisen-
und
Menuettencomponist
von
Johann Philipp Kirnberger.



Berlin, 1767.
Gedruckt und zu finden bey George Ludwig Winter, Königl. privilegirten Buchdrucker.

Tabelle der Würfe zu Polonoisen.

Zum ersten Theile.

mit einem Würfel	1	2	3	4	5	6					
mit zwey Würfeln	2	3	4	5	6	7	8	9	10	11	12
1 Wurf	70	10	42	62	44	72	114	123	131	138	144
2 " "	34	24	6	8	56	30	112	116	147	151	153
3 " "	68	50	60	36	40	4	126	137	143	118	146
4 " "	18	46	2	12	79	28	87	110	113	124	128
5 " "	32	14	52	16	48	22	89	91	101	141	150
6 " "	58	26	66	38	54	64	88	98	115	127	154

Johann Philipp Kirnberger: Der allezeit fertige Polonoisen- und Menuettencomponist, 1767

2 – Algorithmic Composition



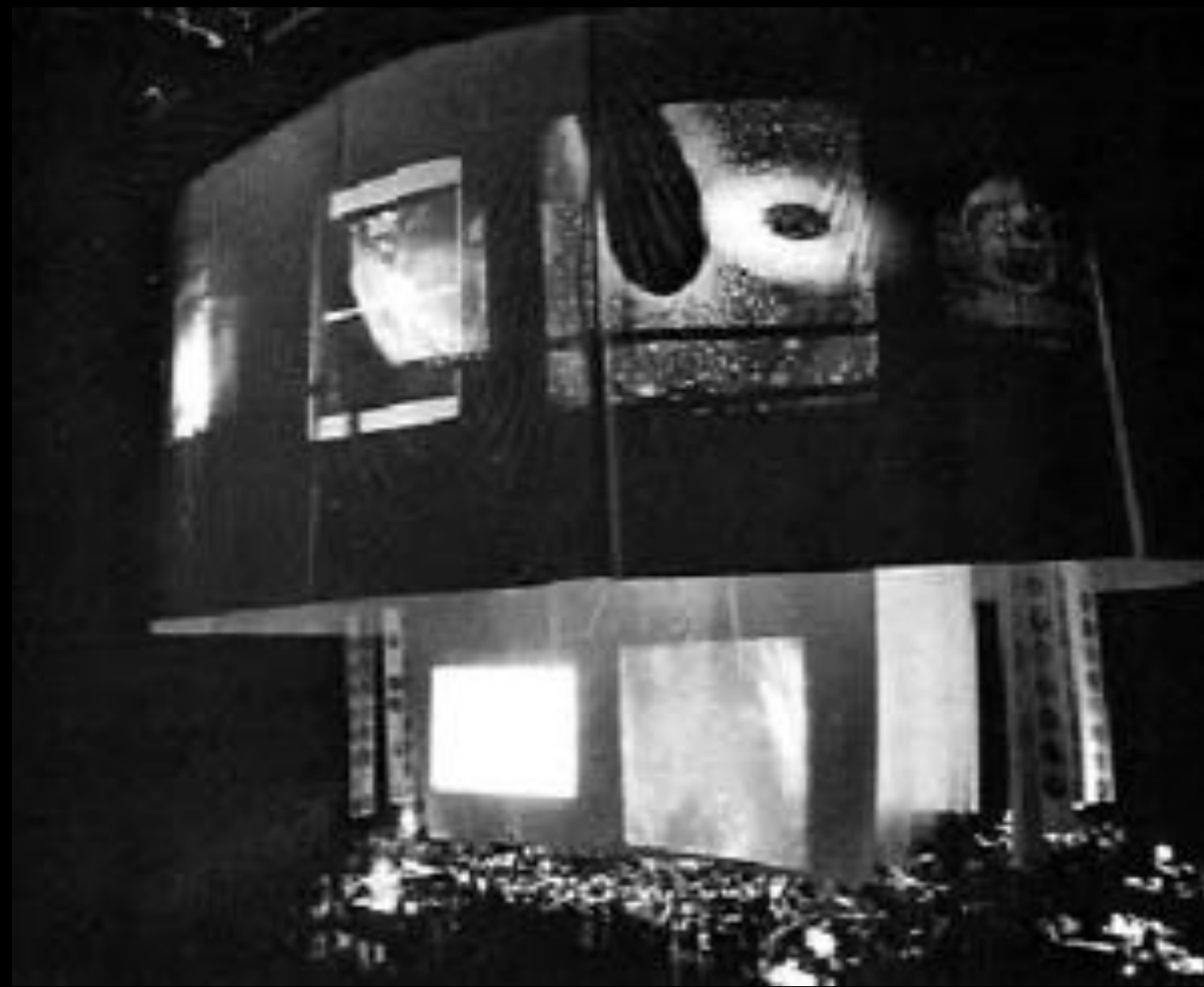
Dietrich Nikolaus Winkel: Componium, 1821

2 – Algorithmic Composition



Joseph Schillinger: The Schillinger System of Musical Composition, 1946

2 – Algorithmic Composition



Lejaren Hiller, 1960 (composer of *Illiac Suite* 1957) and John Cage/Lejaren Hiller: HPSCHD 1969, <https://www.youtube.com/watch?v=SMsDa5Wuguk>

2 – Algorithmic Composition

1982

Création de la norme MIDI et conception à l'université de Stanford par John Chowning du synthétiseur Yamaha DX7 basé sur la synthèse de modulation de fréquence (la synthèse FM).

Invention of the MIDI format and conception by John Chowning at Stanford University of the Yamaha DX7 Synthesizer based on frequency modulation (or FM) synthesis.

Extrait de la partition *Atlantys* (1985) pour 2 synthétiseurs DX7 Yamaha de Tristan Murail (éditions Salabert)

2

ATLANTYS

Synth. I T. Murail

1 Flûte

2 Trompette

3 Violon

4 Violoncelle

5 Basson

6 Clarinette

7 Cor

8 Trombone

9 Tuba

10 Percussion

11 Batterie

12 Piano

13 Orgue

14 Synthétiseur

15 Synthétiseur

16 Synthétiseur

17 Synthétiseur

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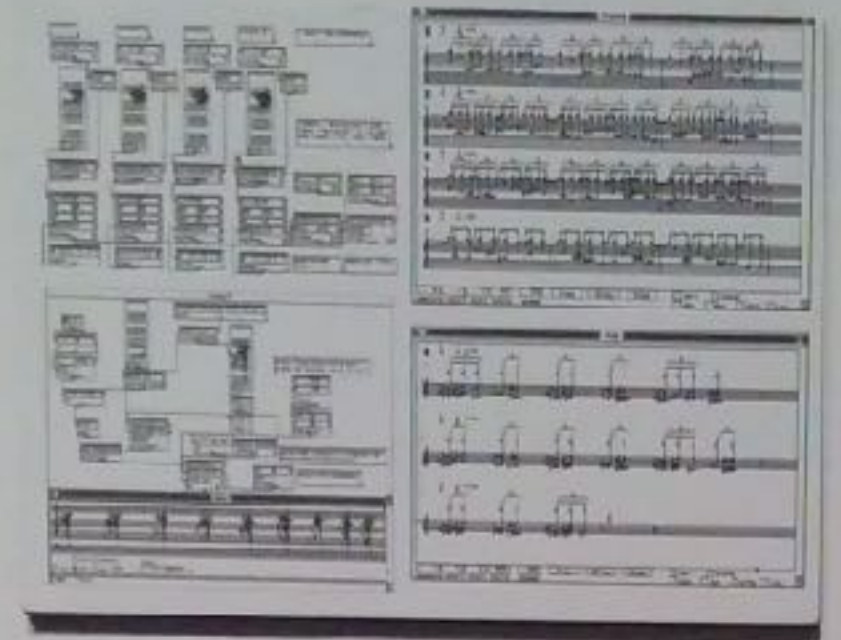
99 Synthétiseur

100 Synthétiseur

1991-94

Conception à l'Ircam de PATCHWORK (par Mikael Laurson, Jacques Duthen et Camilo Rueda), environnement de programmation graphique qui deviendra ensuite PwGL, et de MODALYS (par Francisco Jovino, Jean-Marie Adrien, René Caussé, Joseph Morrison).

Conception at IRCAM of PATCHWORK (by Mikael Laurson, Jacques Duthen and Camilo Rueda), which will become later PwGL and of MODALYS (by Francisco Jovino, Jean-Marie Adrien, René Caussé, Joseph Morrison).



Une série de patches en Patchwork (source : Gérard Assayag)

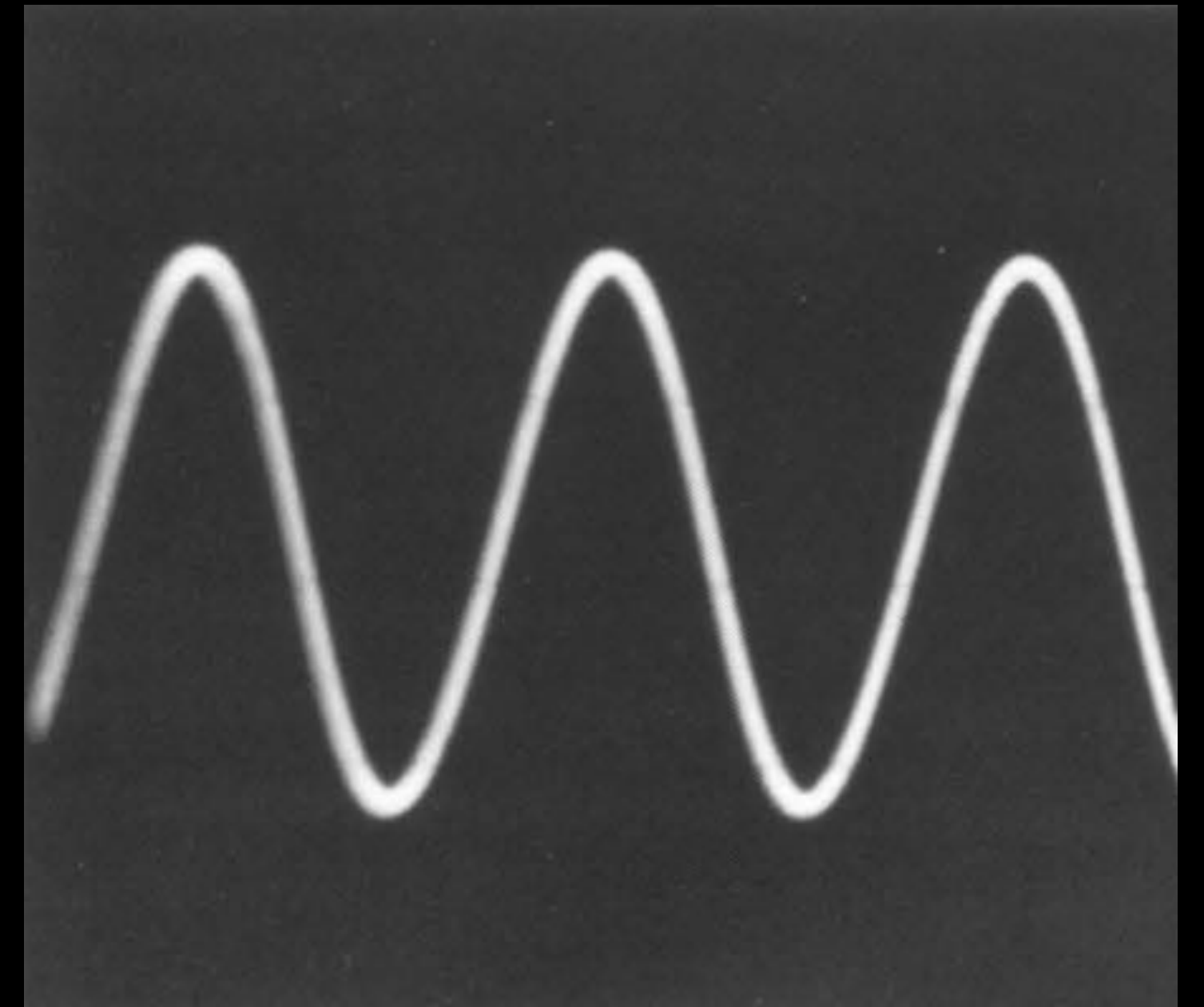
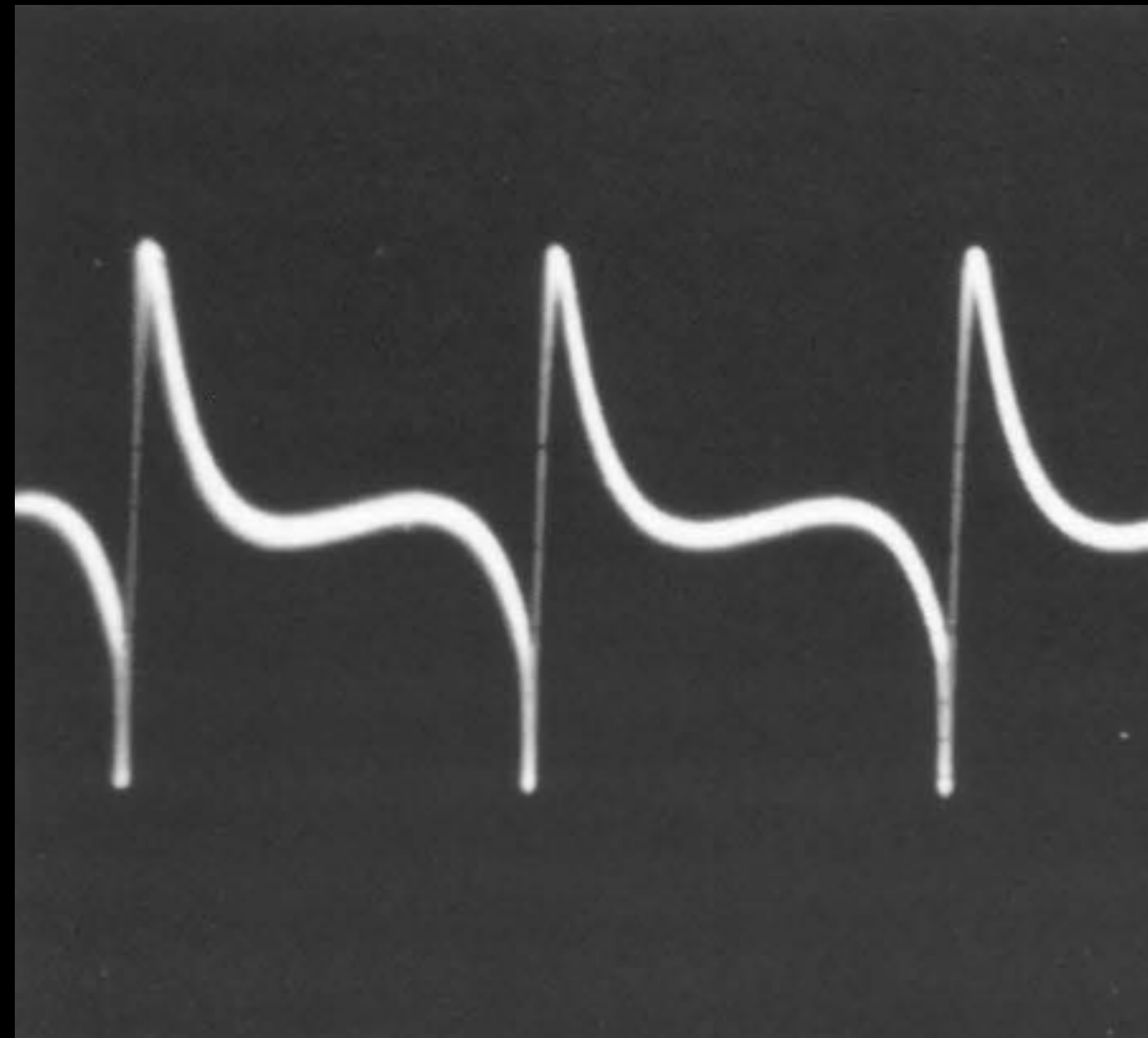
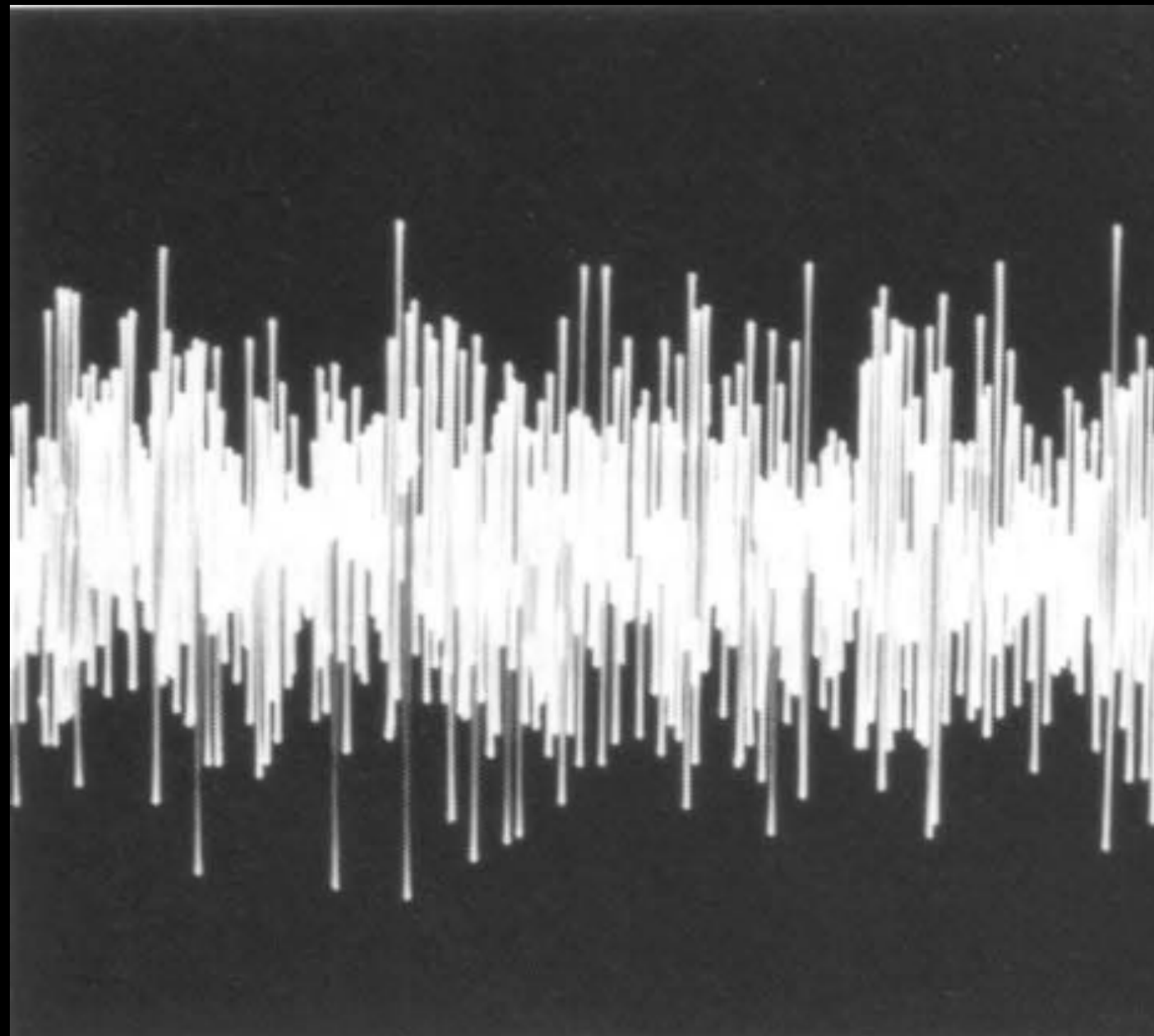
1994-96

Conception d'OPENMUSIC (par Gérard Assayag et Carlos Agon) à l'Ircam et de SUPERCOLLIDER (par James McCartney).

Conception of OPENMUSIC (by Gérard Assayag and Carlos Agon) at IRCAM and of SUPERCOLLIDER (by James McCartney).

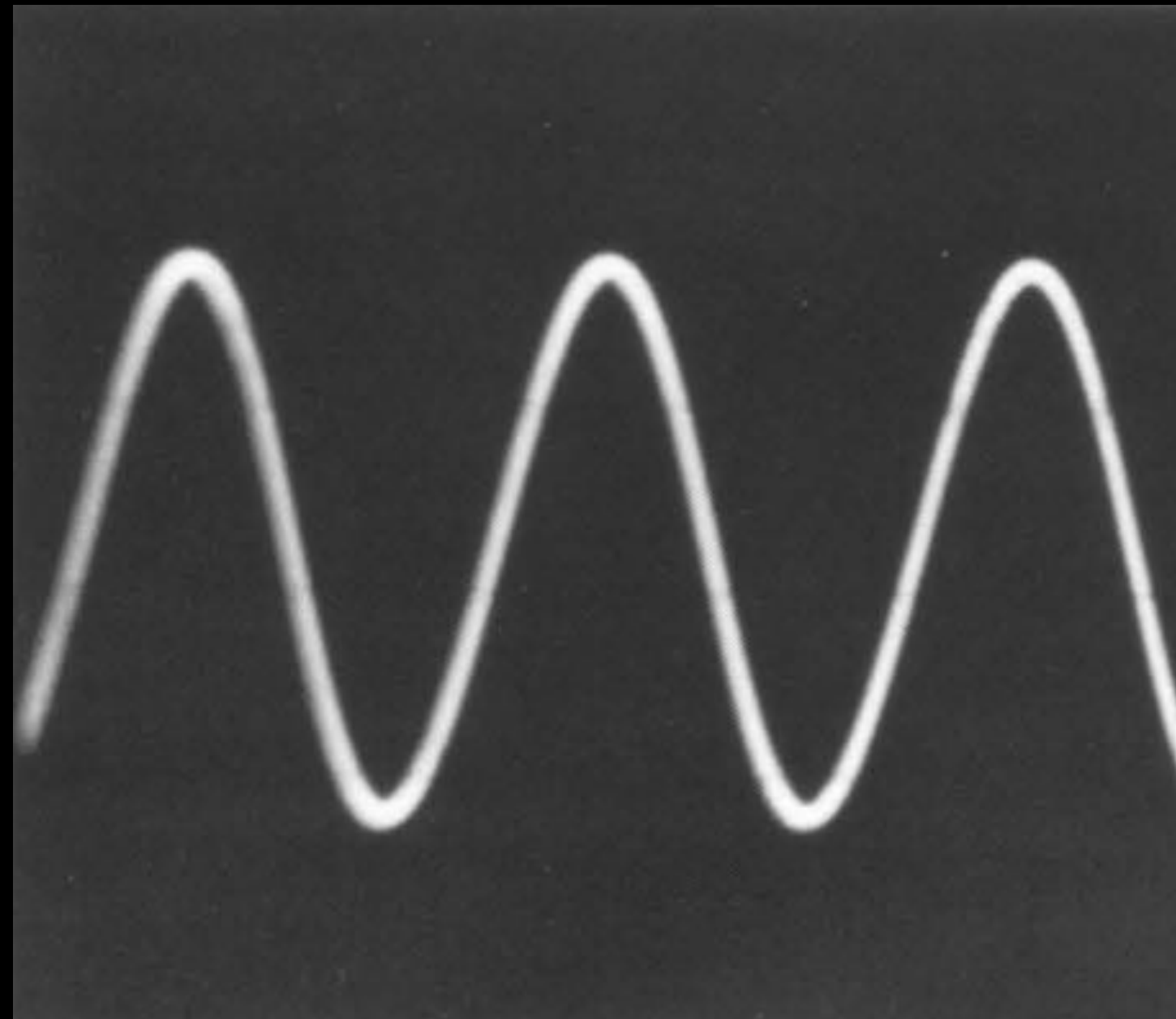
Overview developments of 1980s and 90s (MIDI 1982, Visual Programming 1990er Jahre, Photos from the Exhibition Coder le Monde 2018)

3 – Synthesized Sound



Taylor, Charles: „The Physics of Sound“, in: Kruth, Patricia und Henry Stobart (Hrsg.): *Sound*, Cambridge: Cambridge University Press 2000, S. 34–64, hier S. 36.

3 – Synthesized Sound

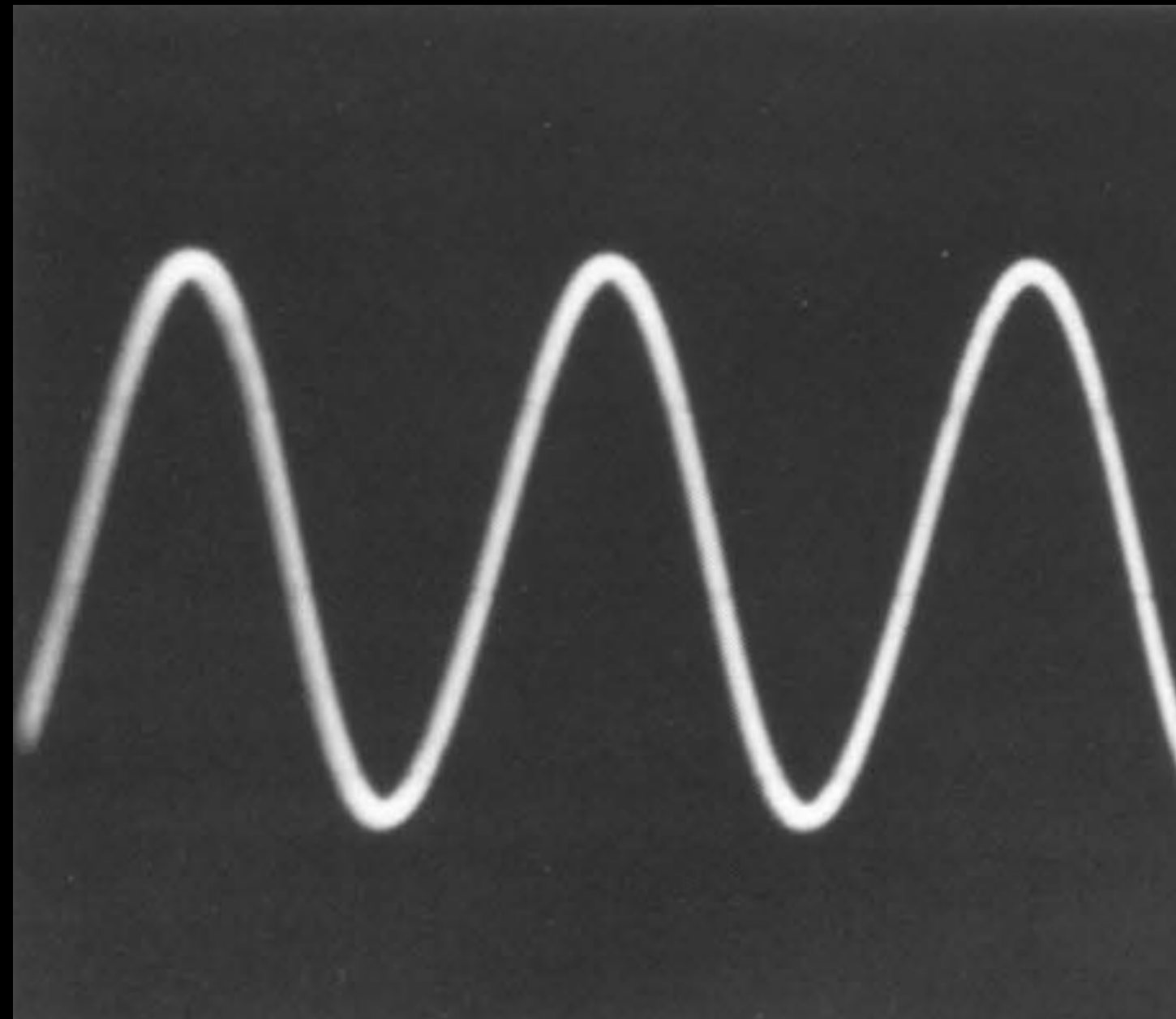


“‘[S]ynthesis’ did not necessarily refer to an **electronic** or even **mechanical** process or device; it could be **mathematical** or **philosophical**. But with the crystallization of **synthesizer instruments in the postwar period**, all three of these connotations would remain present: a synthesizer was necessarily an electronic instrument which, if not properly mechanical, was expected to **automate tasks in sound production or music composition** [...]”

Rodgers, Tara: „Synthesizing Sound: Metaphor in Audio-Technical Discourse and Synthesis History“, PhD, Montréal: McGill University, Montreal 2010, S. 149.

Taylor, Charles: „The Physics of Sound“, in: Kruth, Patricia und Henry Stobart (Hrsg.): *Sound*, Cambridge: Cambridge University Press 2000, S. 34–64, hier S. 36.

3 – Synthesized Sound

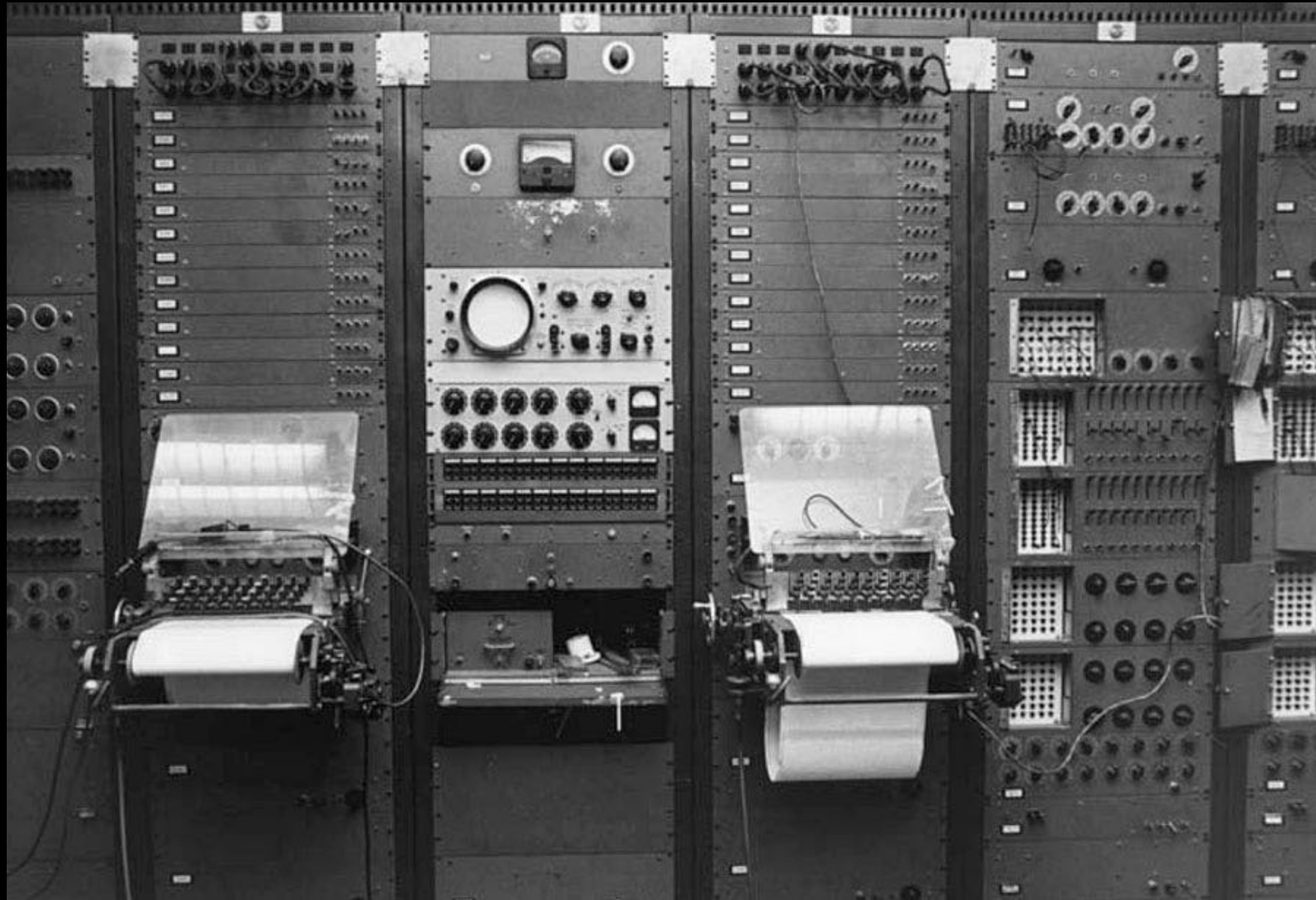


“Visualizing sound [...] was a central task of the new science of acoustics [...]. Sound had, according to the accepted techniques of science, to be seen in order to be quantified, measured, and recorded [which] required the simultaneous construction of sound [...].”

Jonathan Sterne zit. n. Rodgers, Tara: „Synthesizing Sound: Metaphor in Audio-Technical Discourse and Synthesis History“, PhD, Montréal: McGill University, Montreal 2010, S. 155.

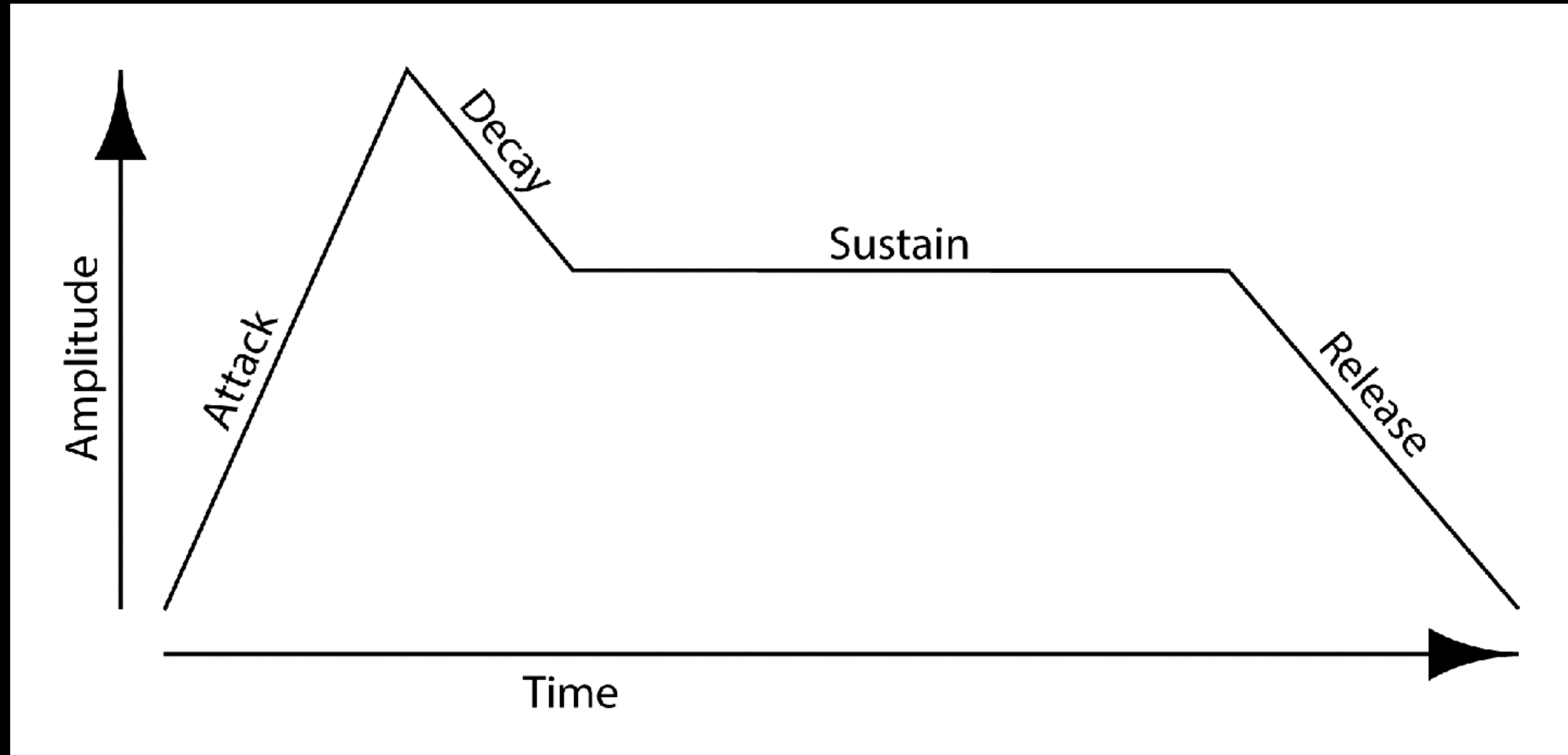
Taylor, Charles: „The Physics of Sound“, in: Kruth, Patricia und Henry Stobart (Hrsg.): *Sound*, Cambridge: Cambridge University Press 2000, S. 34–64, hier S. 36.

3 – Synthesized Sound



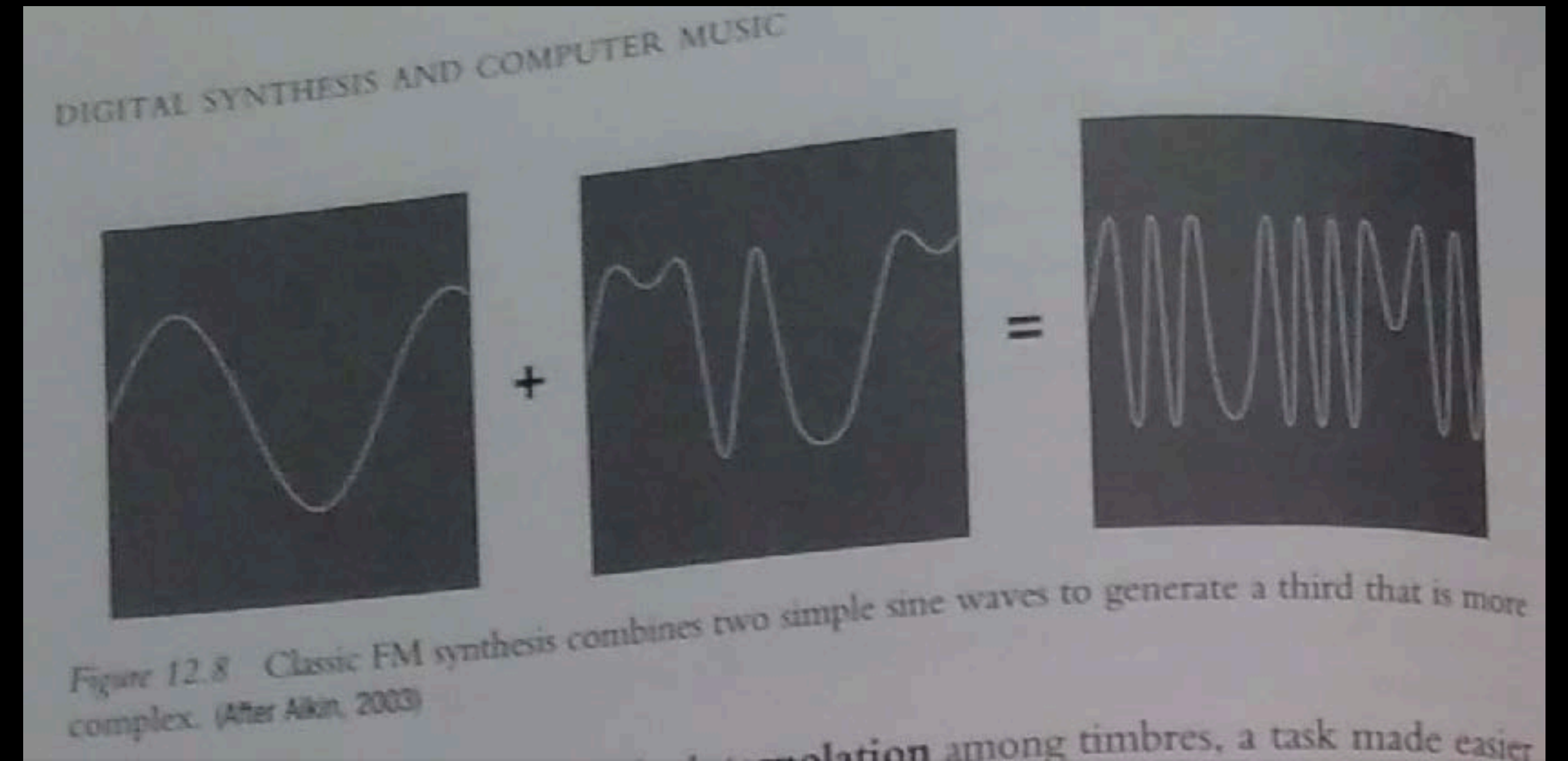
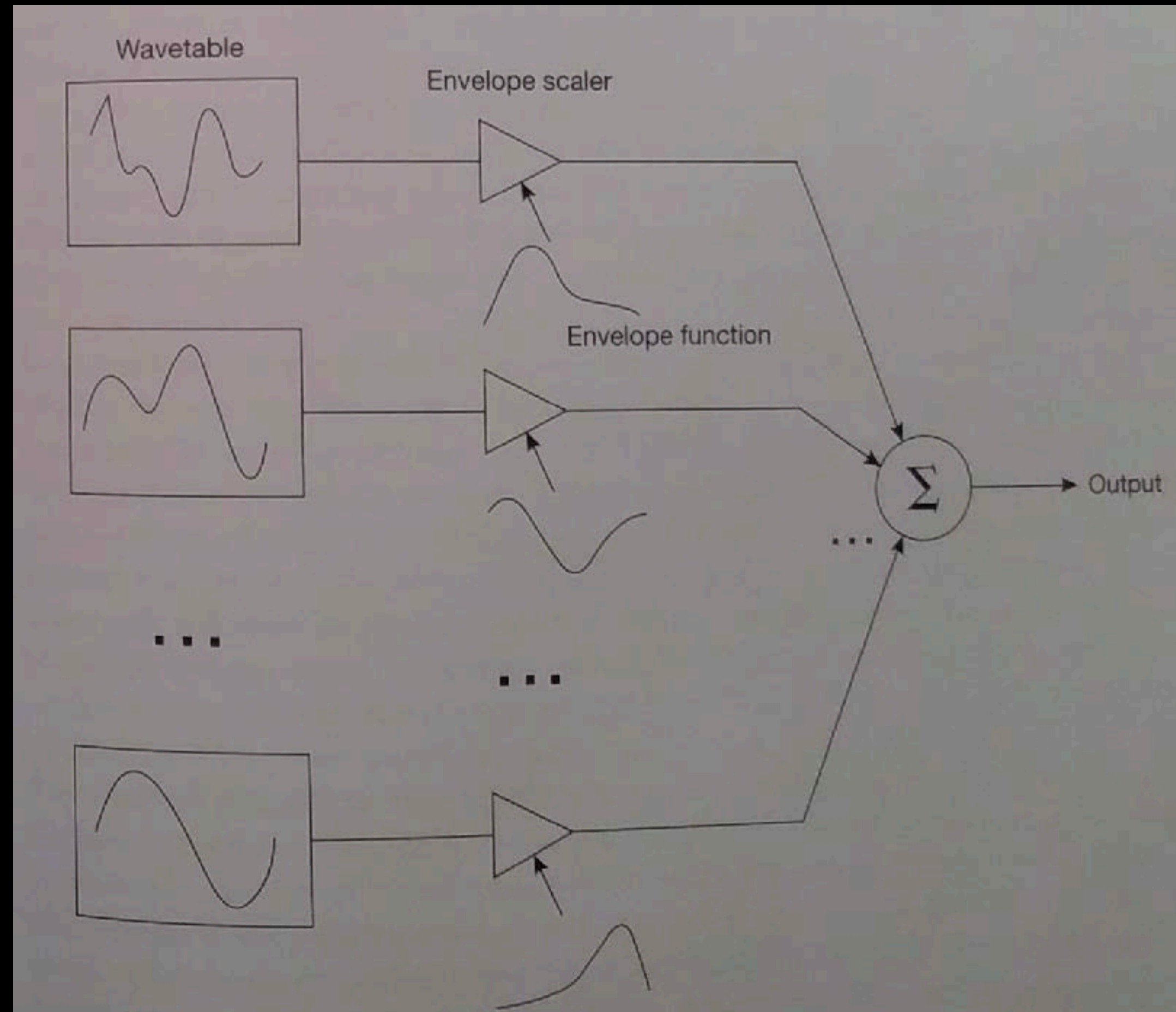
RCA Mark II Electronic Music Synthesizer, 1958

3 – Synthesized Sound



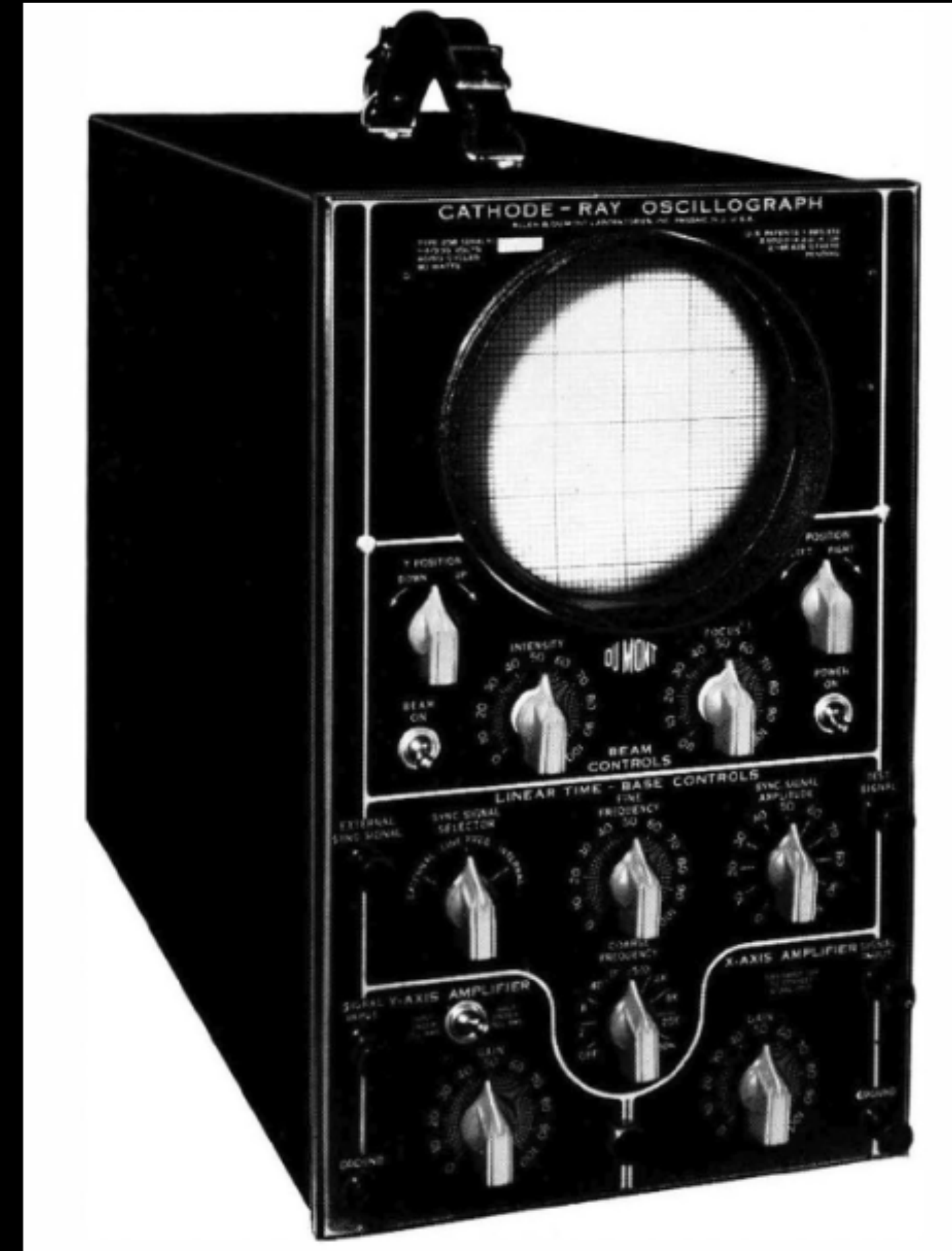
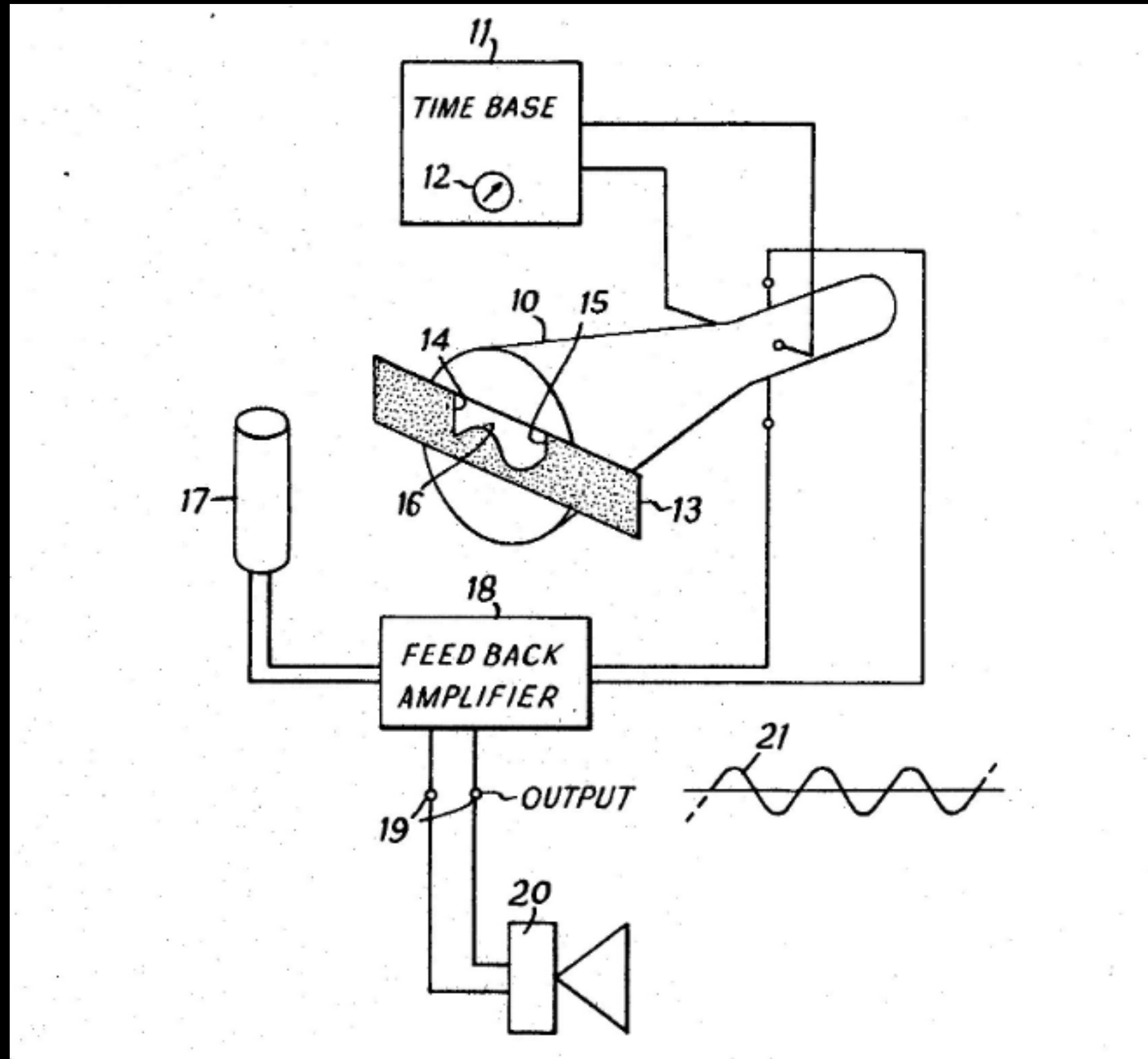
Hüllkurve nach Holmes, Thom: Electronic and Experimental Music. Technology, Music, and Culture, 3. Aufl., New York: Routledge 2008, S. 196.

3 – Synthesized Sound



Holmes, Thom: Electronic and Experimental Music. Technology, Music, and Culture, 3. Aufl., New York: Routledge 2008, S. 305f.

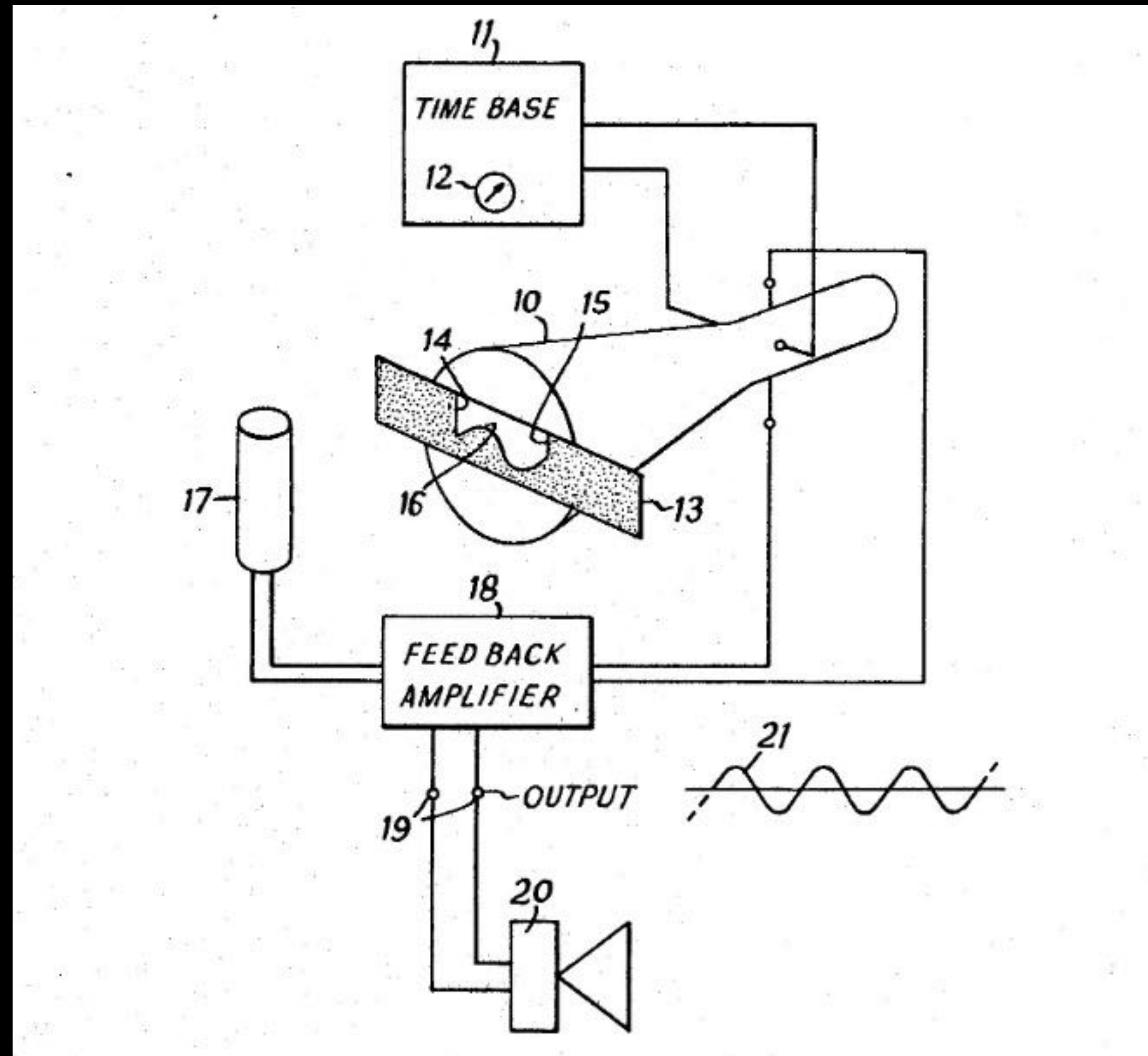
3 – Synthesized Sound



Daphne Oram: Waveform Generators, patent U.S. 3,478,792A, 18.11.1969, Fig. 1.

Allen B. Du Mont Laboratories: Du Mont Type 208 Cathode-Ray Oscilloscope, product info sheet, ca. 1941.

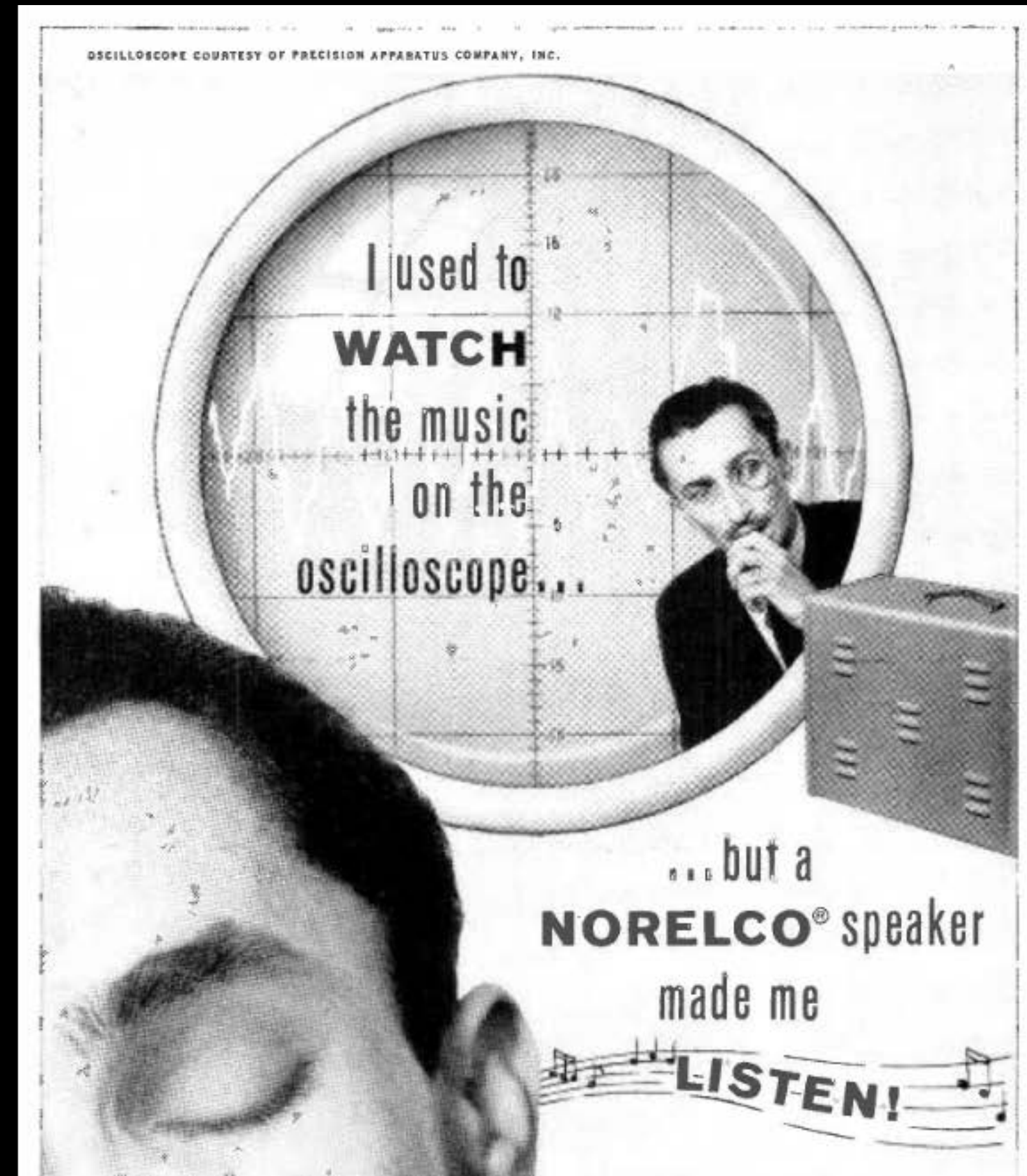
3 – Synthesized Sound



“Helmholtz’s claim that the stationary part of a sound could suffice to describe Klangfarbe. [...] Arnold Schoenberg in turn used this effect to merge the sounds of musical instruments into new orchestral colors.”

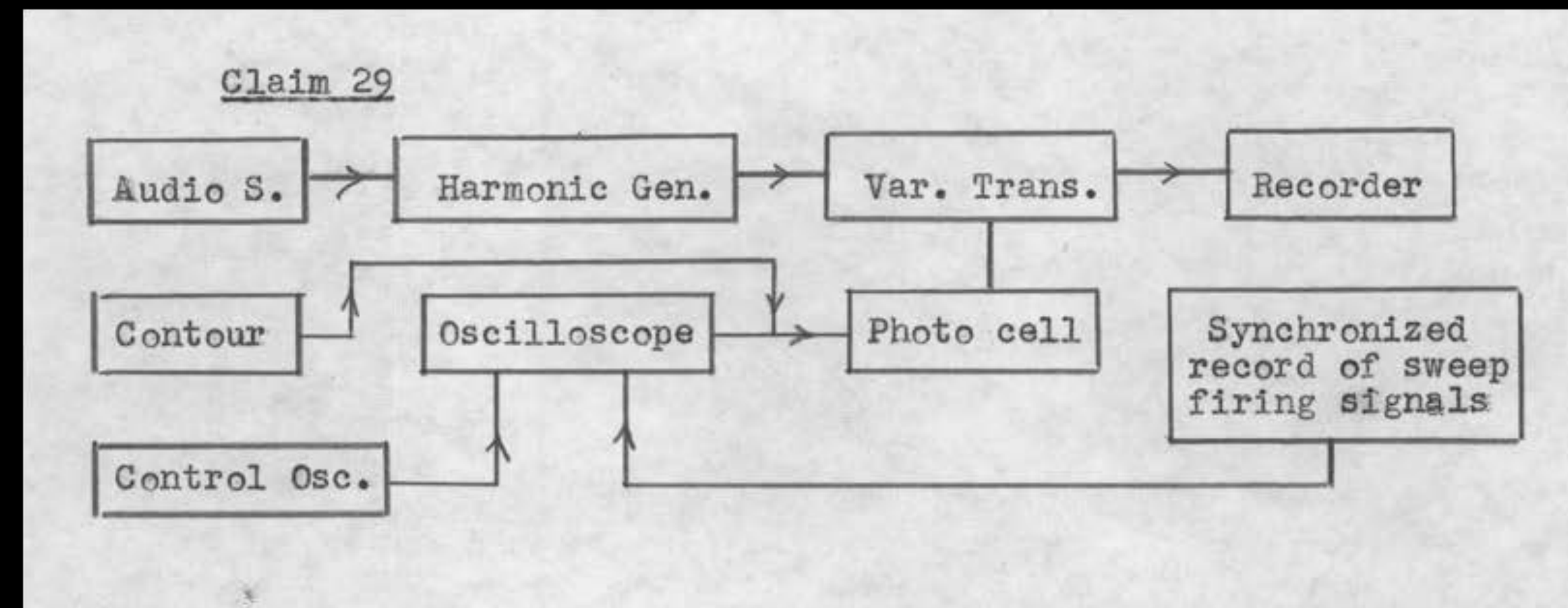
Julia Kursell: “Experiments on Tone Color in Music and Acoustics: Helmholtz, Schoenberg, and Klangfarbenmelodie”, *Osiris* 28/1 (2013), pp. 191–211, here pp. 191, 207.

3 – Synthesized Sound



Alan Phillips: „Osmond Kendall’s Marvelous Music Machine“, *Macleans* (11 June 1955), here p. 22.
Ad for speakers, *Audio & Audio Engineering* (May 1958), here p. 55.

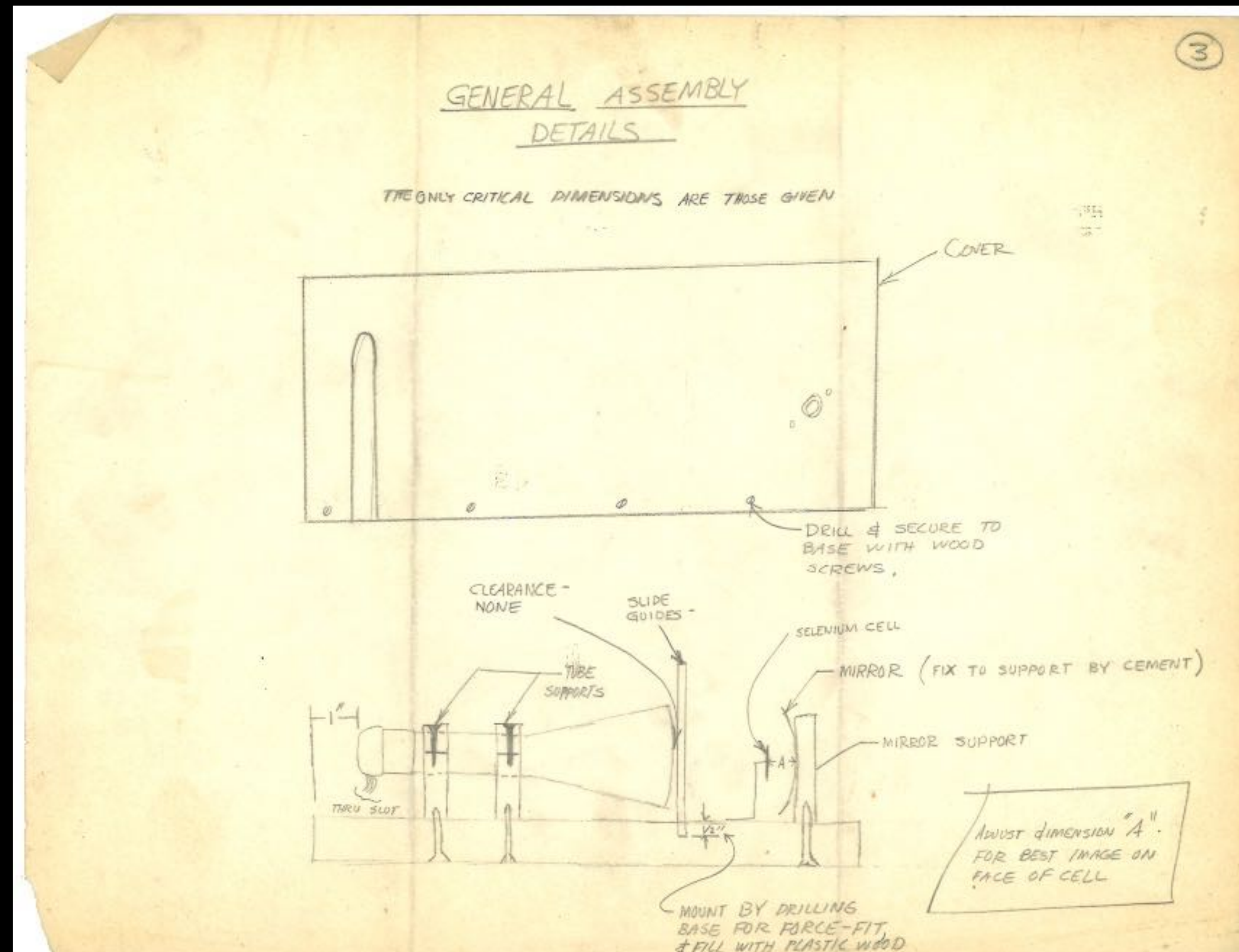
3 – Synthesized Sound



Alan Phillips: „Osmond Kendall’s Marvelous Music Machine“, *Macleans* (11 June 1955), here p. 22.

Osmond K. Kendall: „The Kendall Composer-Tron“, Draft for a Patent Application, ca. 1951, Department of External Affairs Fonds, RG25 8030, Folder 405-AD-40, Library Archives of Canada, Ottawa.

3 – Synthesized Sound



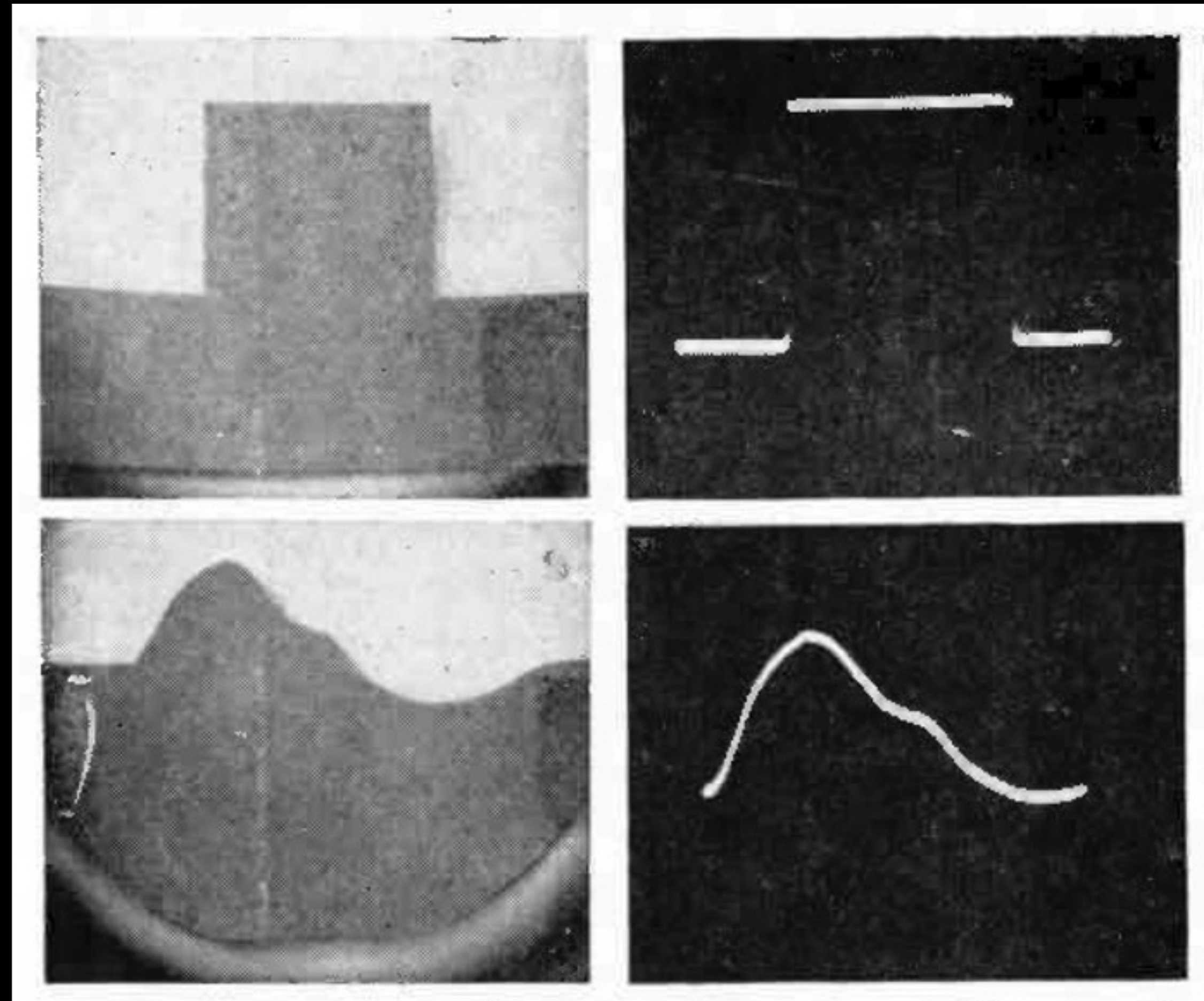
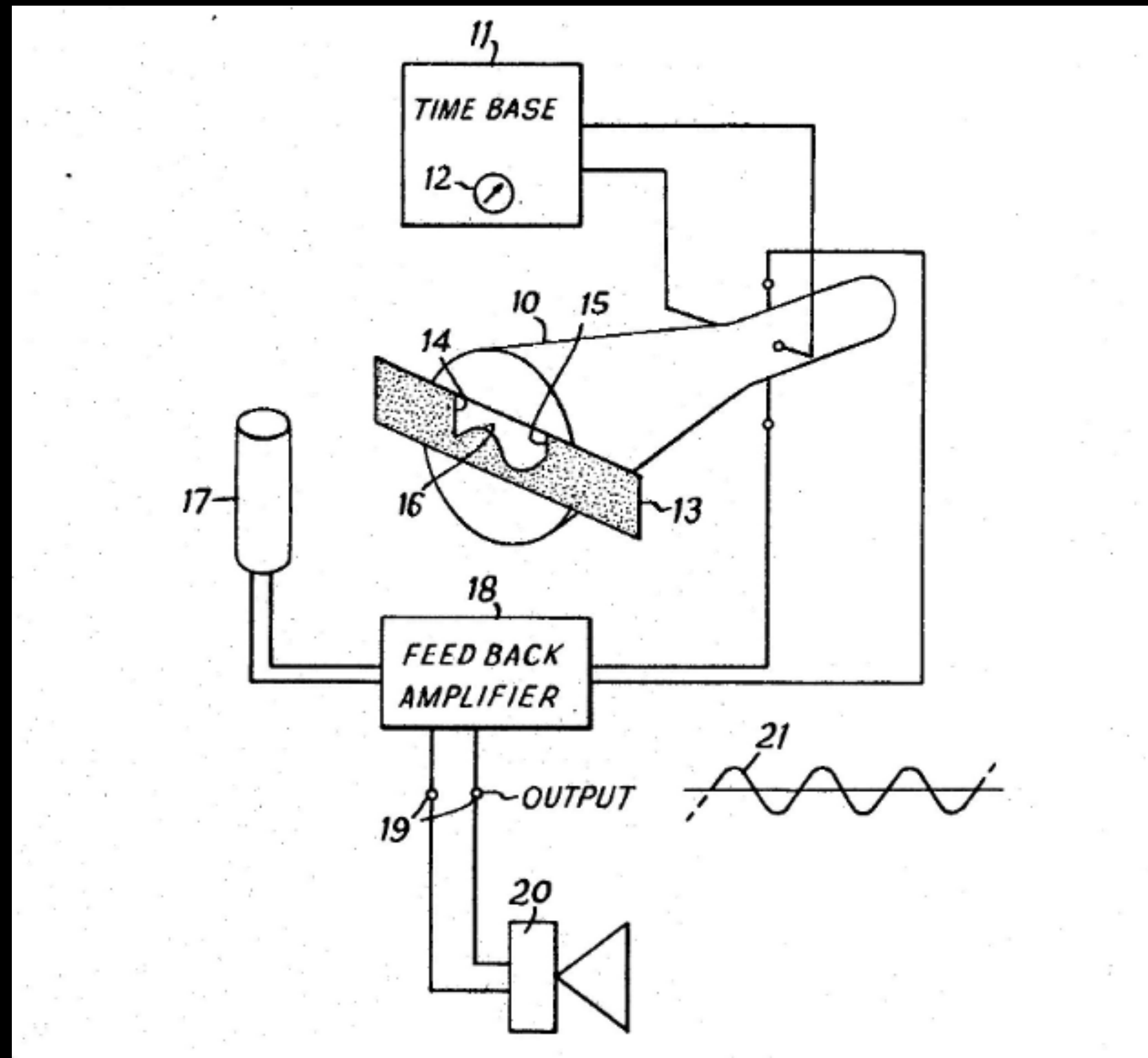
Max Brand, Richard W. Rodgers: Sketch for assembling the optical unit (envelope shaping resource assembly), side view, 1957-59. Elisabeth Schimana (ed.): *Maschinen Oper: Der Komponist Max Brand. Visionen, Brüche und die Realität*, Wien: Hollitzer Wissenschaftsverlag 2016, p. 118.

3 – Synthesized Sound



Daphne Oram, Graham Wrench: Oramics timbre scanning unit and stencils, 1964-66, Science Museum London.

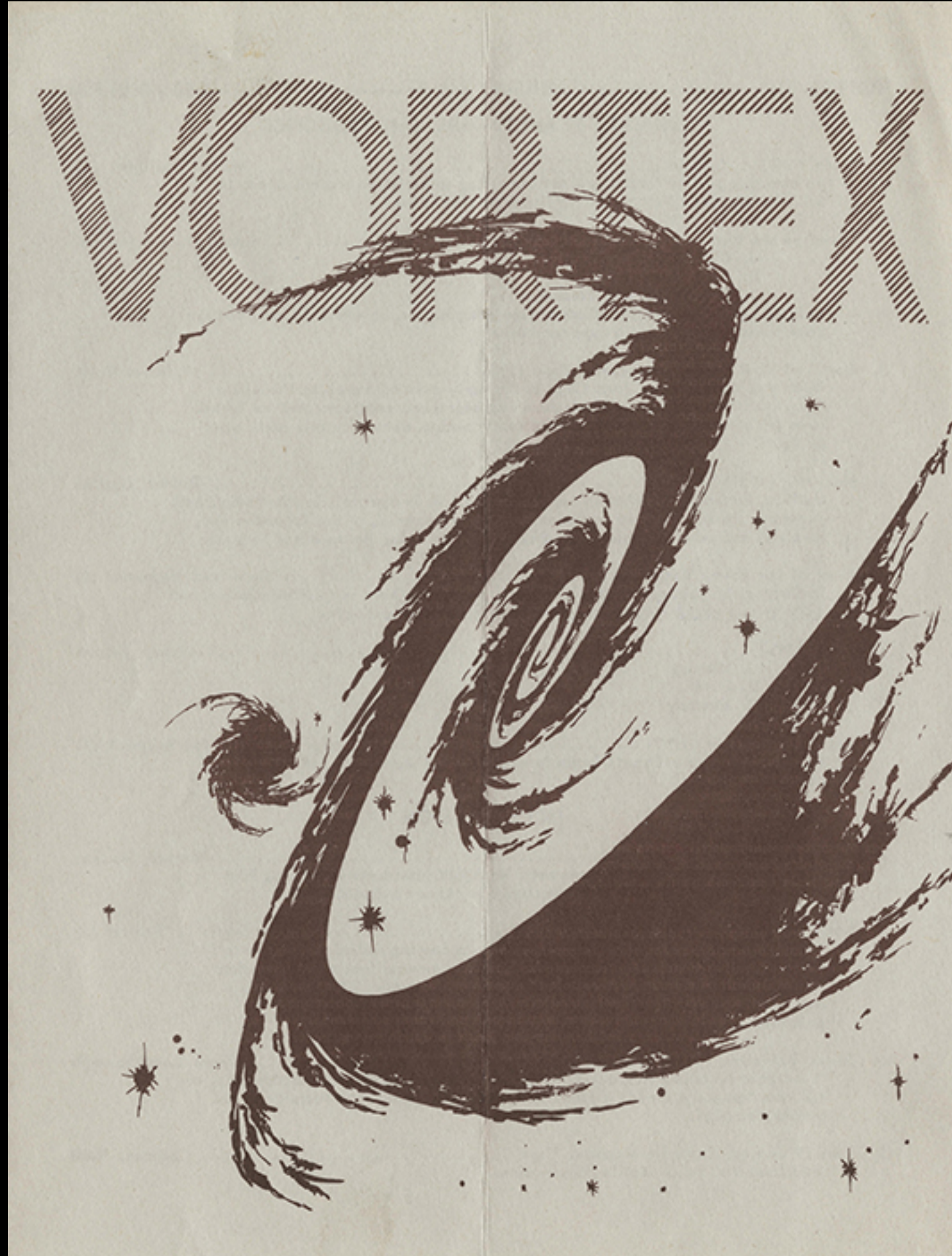
3 – Synthesized Sound



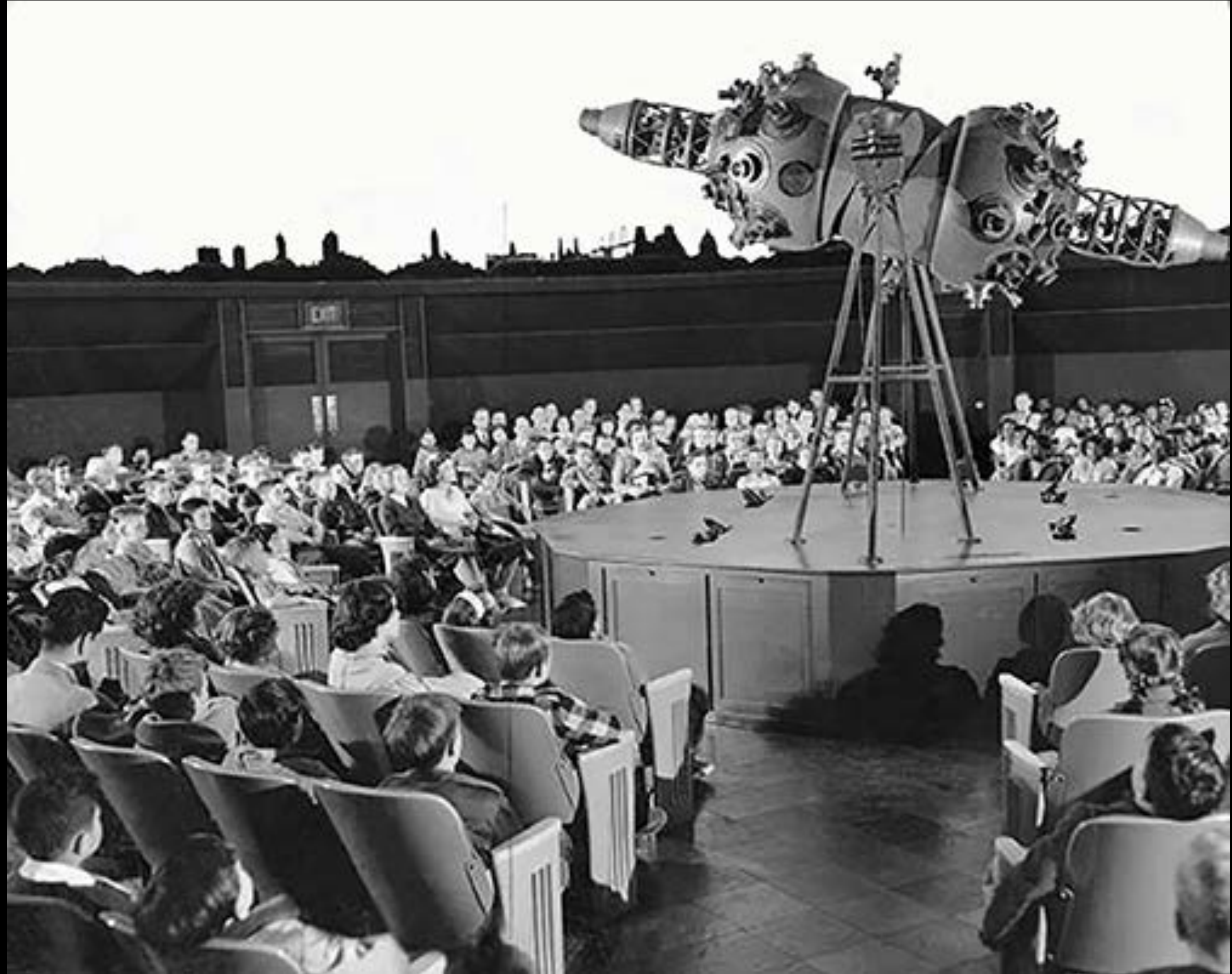
Daphne Oram: Waveform Generators, patent U.S. 3,478,792A, 18.11.1969, Fig. 1.

David E. Sunstein: „Photoelectric Waveform Generator“, *Electronics* (February 1949), p. 101.

4 – Visual Music



Vortex program flyer, 1958

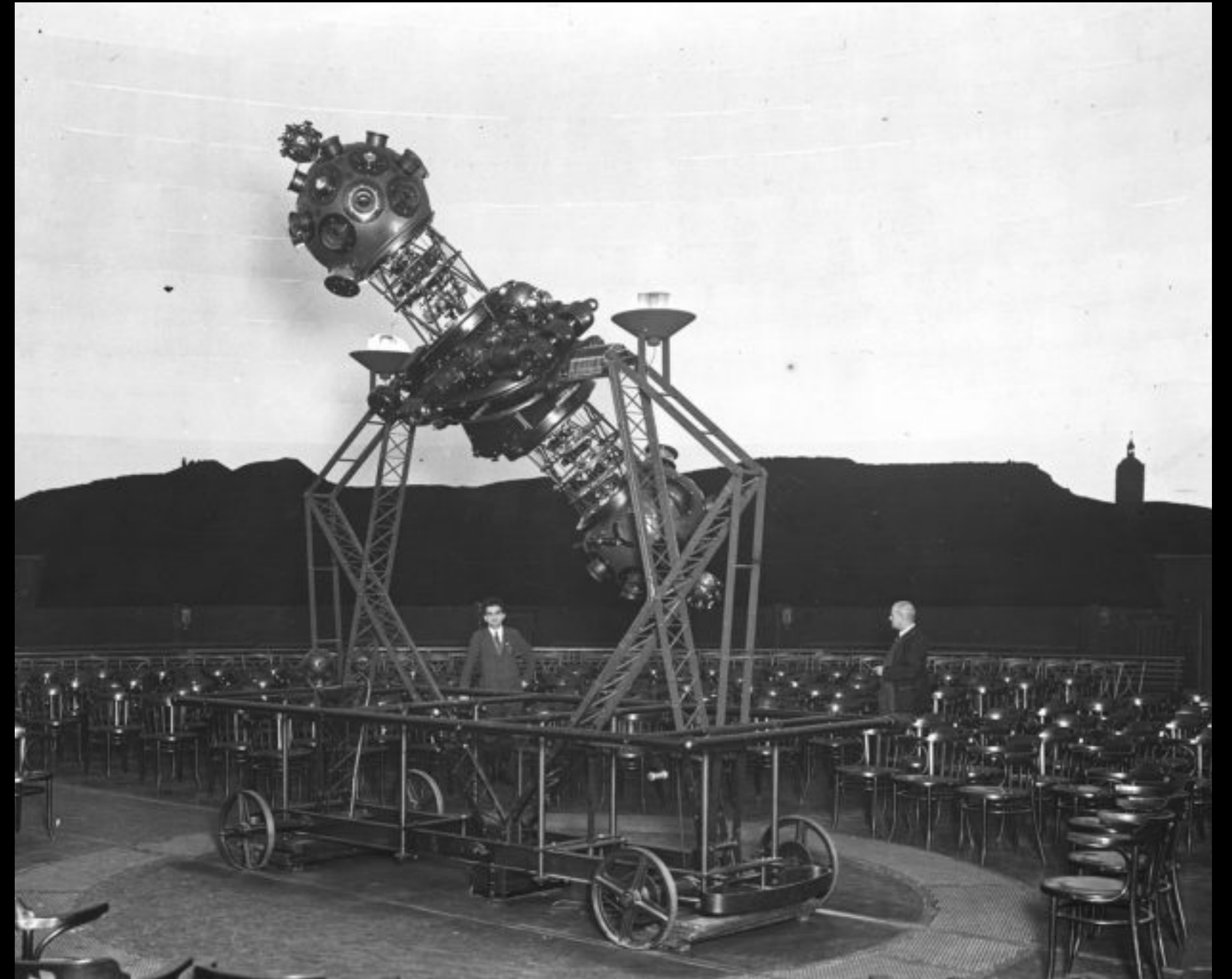


Morrison-Planetarium, San Francisco

4 – Visual Music

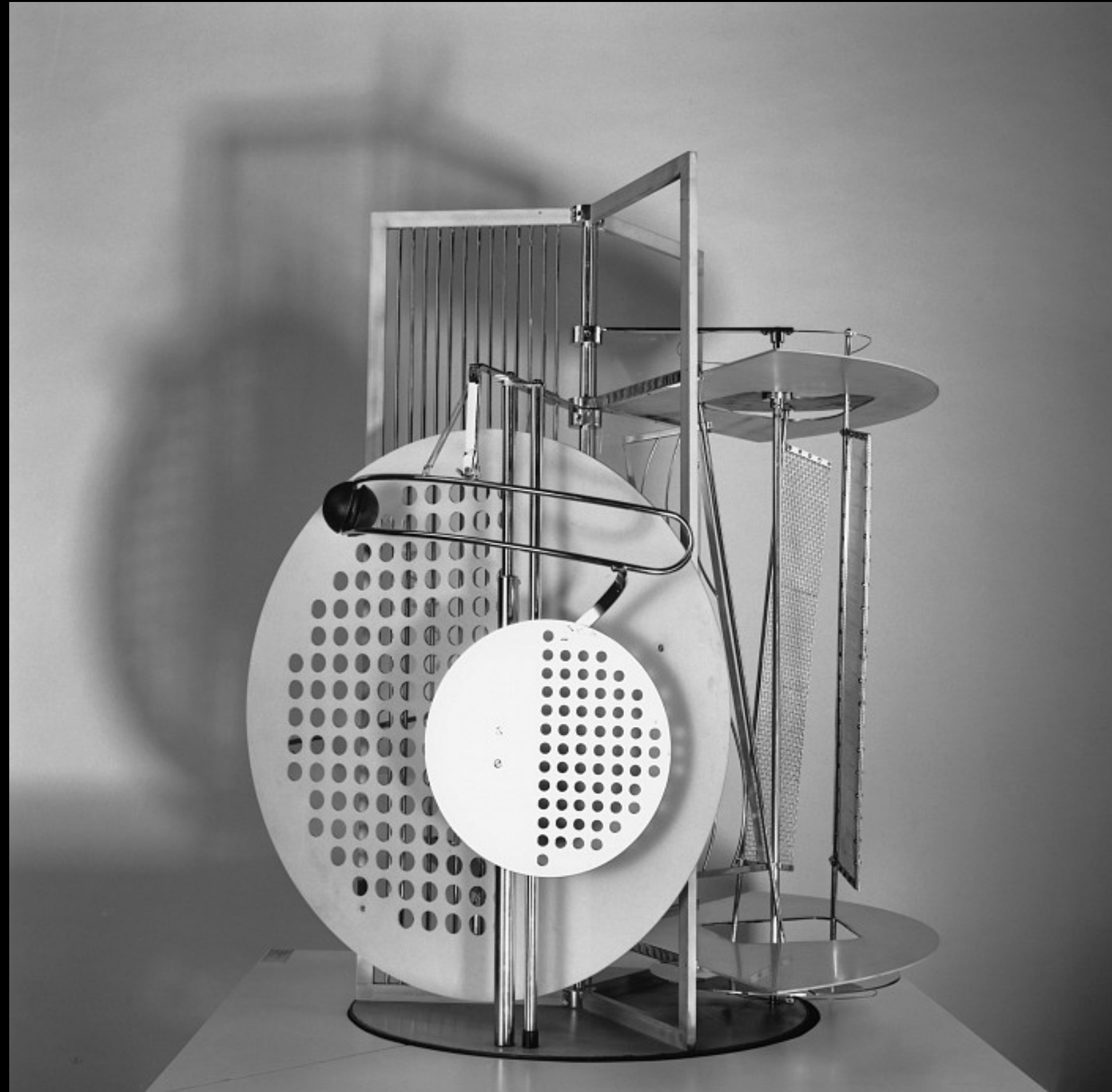


Zeiss Planetarium Jena during construction, 1925



Zeiss star projector, 1926

4 – Visual Music

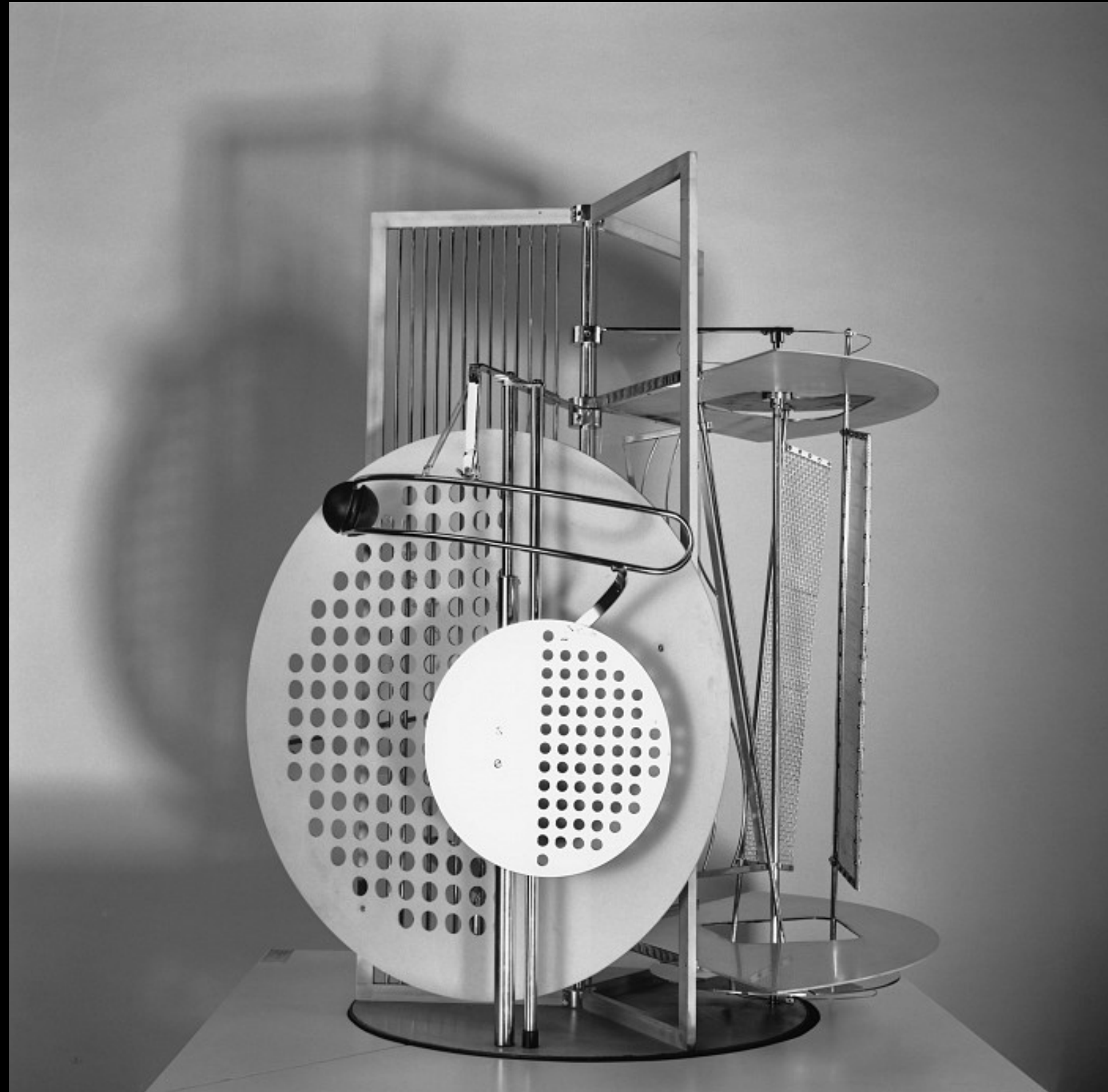


László Moholy-Nagy's light prop, 1930
(Lichtrequisit einer elektrischen Bühne,
reconstruction 1970)



Excerpt from Lichtspiel Schwarz-Weiss-Grau, 1930
16mm, b/w, silent, 5'49"

4 – Visual Music

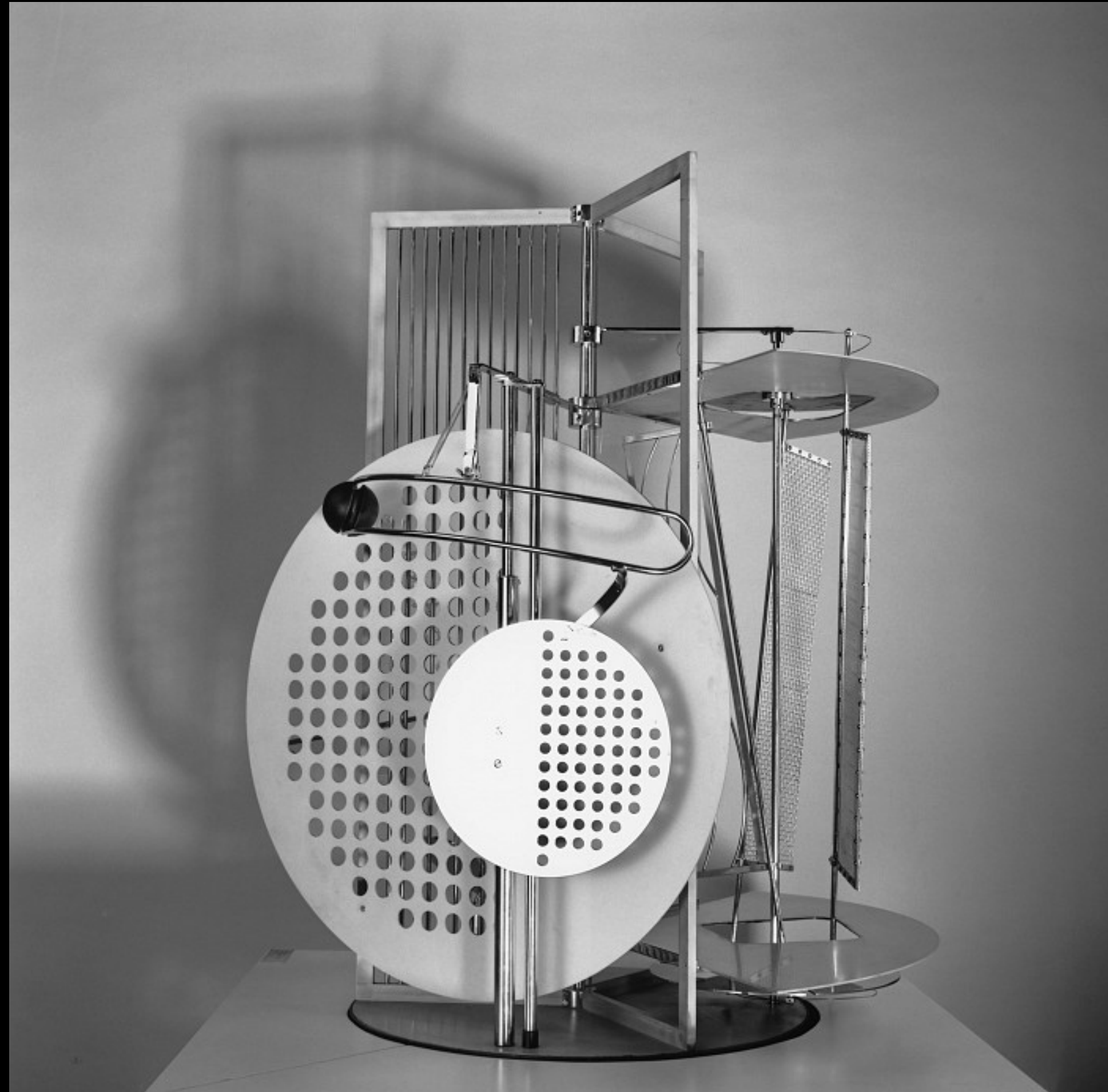


László Moholy-Nagy's light prop, 1930
(Lichtrequisit einer elektrischen Bühne,
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„mit direktem licht, mit fließendem, oszillierendem, farbigem licht ‚malen‘. die heute existierenden optischen instrumente, die regulierbaren künstlichen (elektrischen) lichtquellen bieten eine nicht zu unterschätzende handhabe dafür, auch wenn sie vorläufig nur durch zufall oder auf umwegen – meist als lichtreklame oder teaterrequisit – in die hände des gestalters fallen.“

Moholy-Nagy, László: Von Material zu Architektur, p. 91.

4 – Visual Music



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Moholy-Nagy, László: Von Material zu Architektur, p. 91

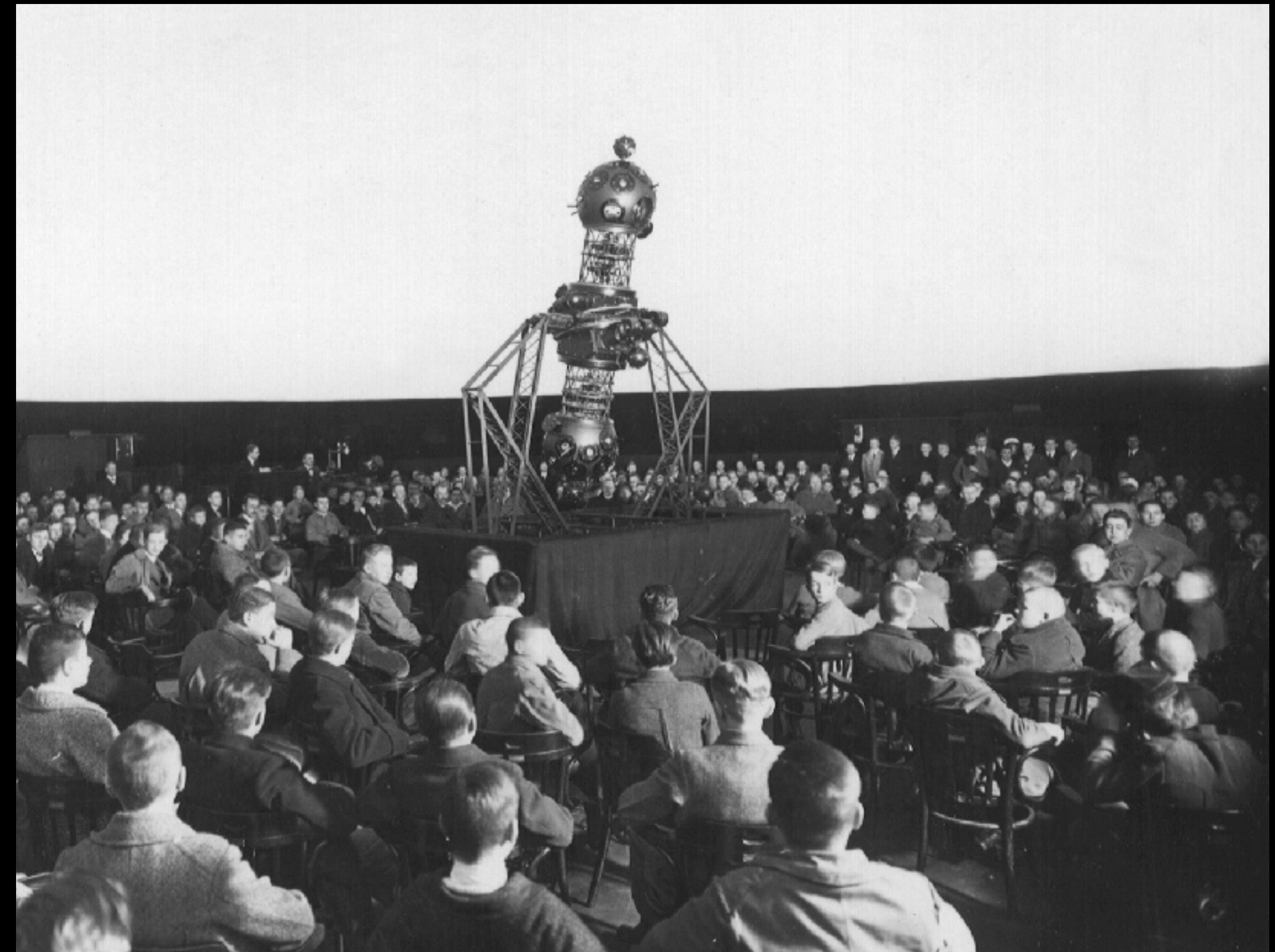
Sie haben eine Projektionsästhetik gemeinsam,
„die mit dem Maschinenkult der Moderne
verknüpft ist.“

Williams, Alena J.: „Akte der Enthüllung. Das Planetarium und László Moholy-Nagys Kunst der Projektion“, in: Beyer, Andreas und Guillaume Cassegrain (Hrsg.): *Mouvement. Bewegung. Über die dynamischen Potenziale der Kunst*, Berlin: Deutscher Kunstverlag 2015, pp. 181–201, here p. 193.

4 – Visual Music



Zeiss Planetarium Jena during construction, 1925

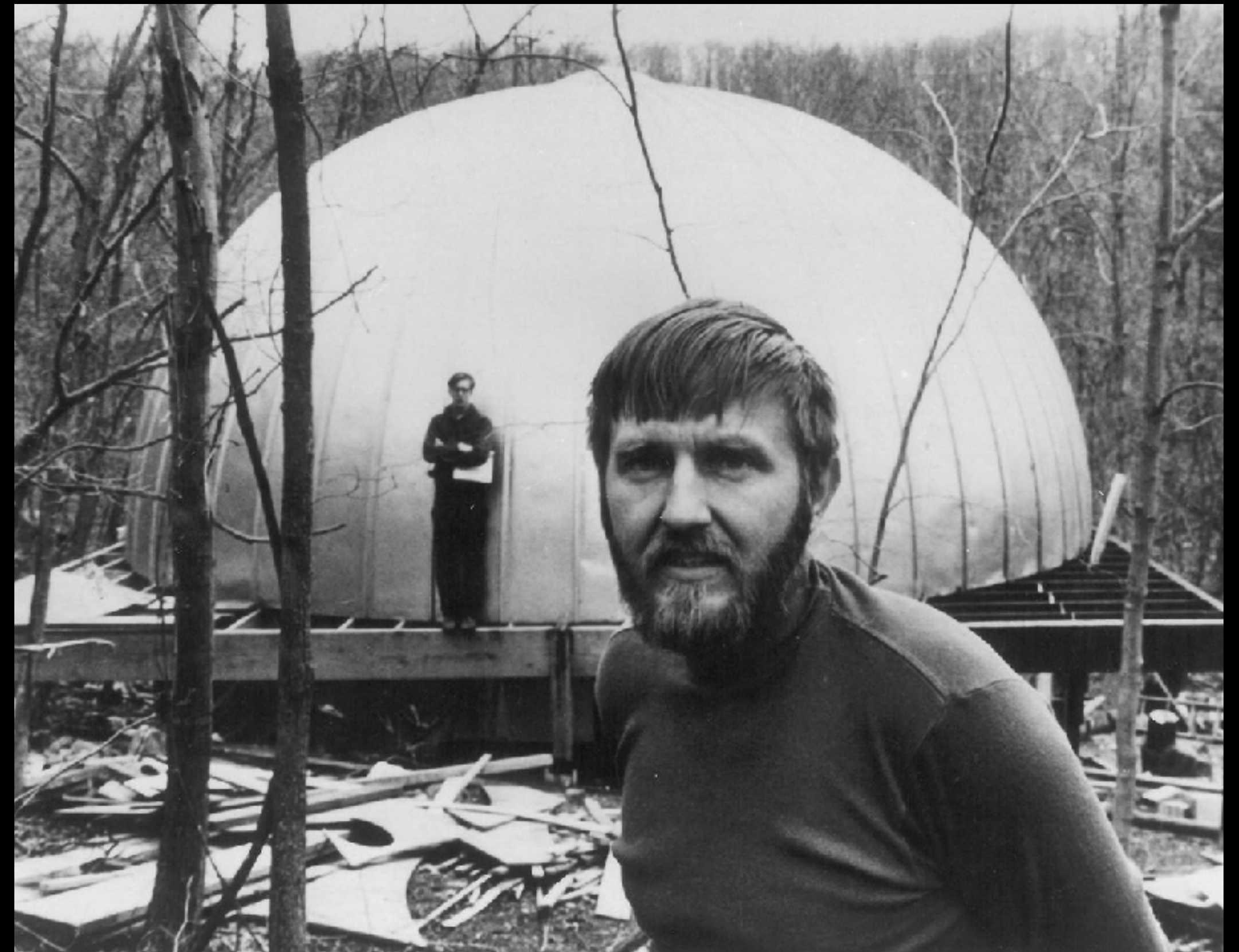


Zeiss star projector, 1926

4 – Visual Music

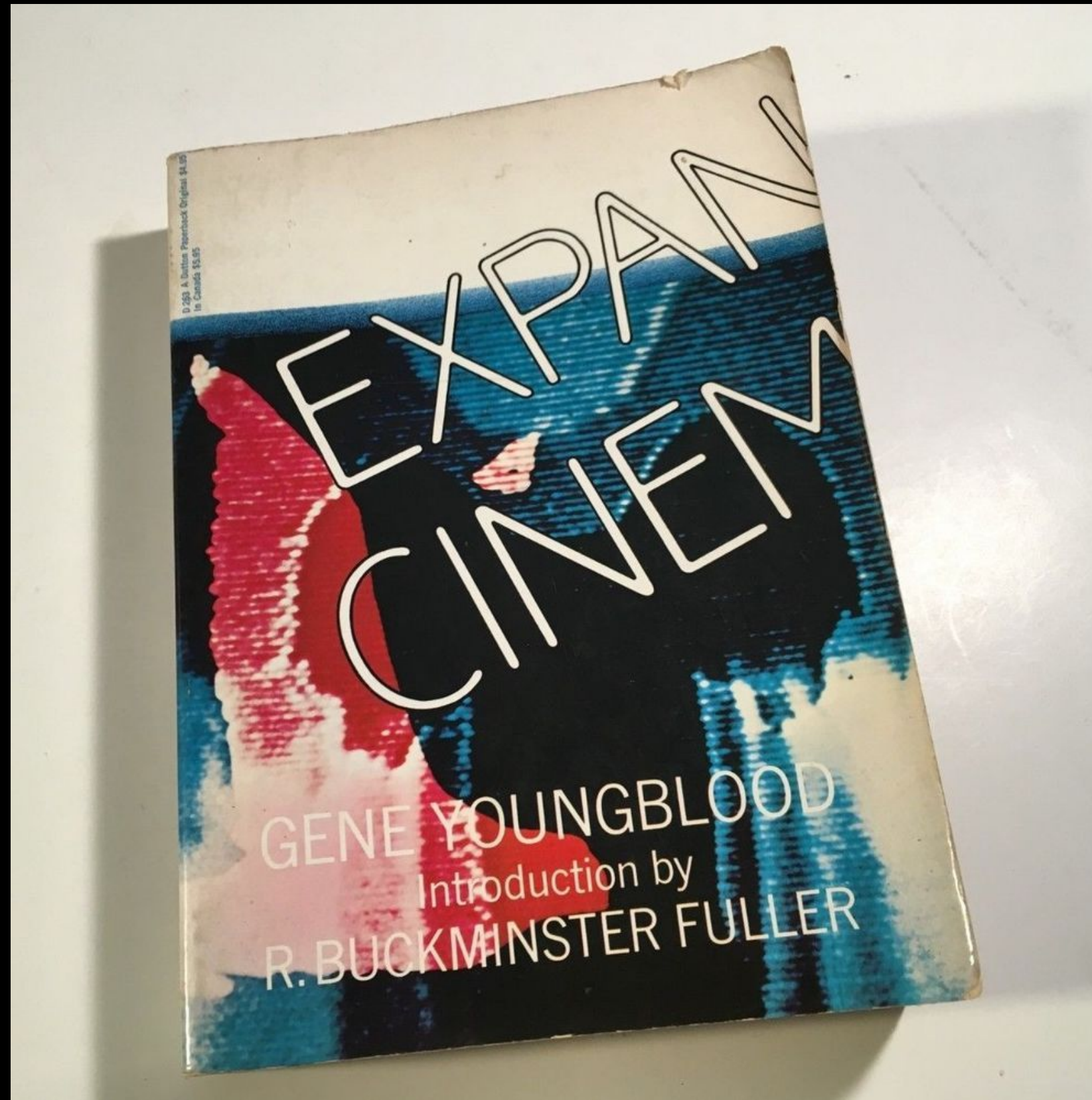


Zeiss Planetarium Jena during construction, 1925



VanDerBeek's Movie-Drome, 1966

4 – Visual Music

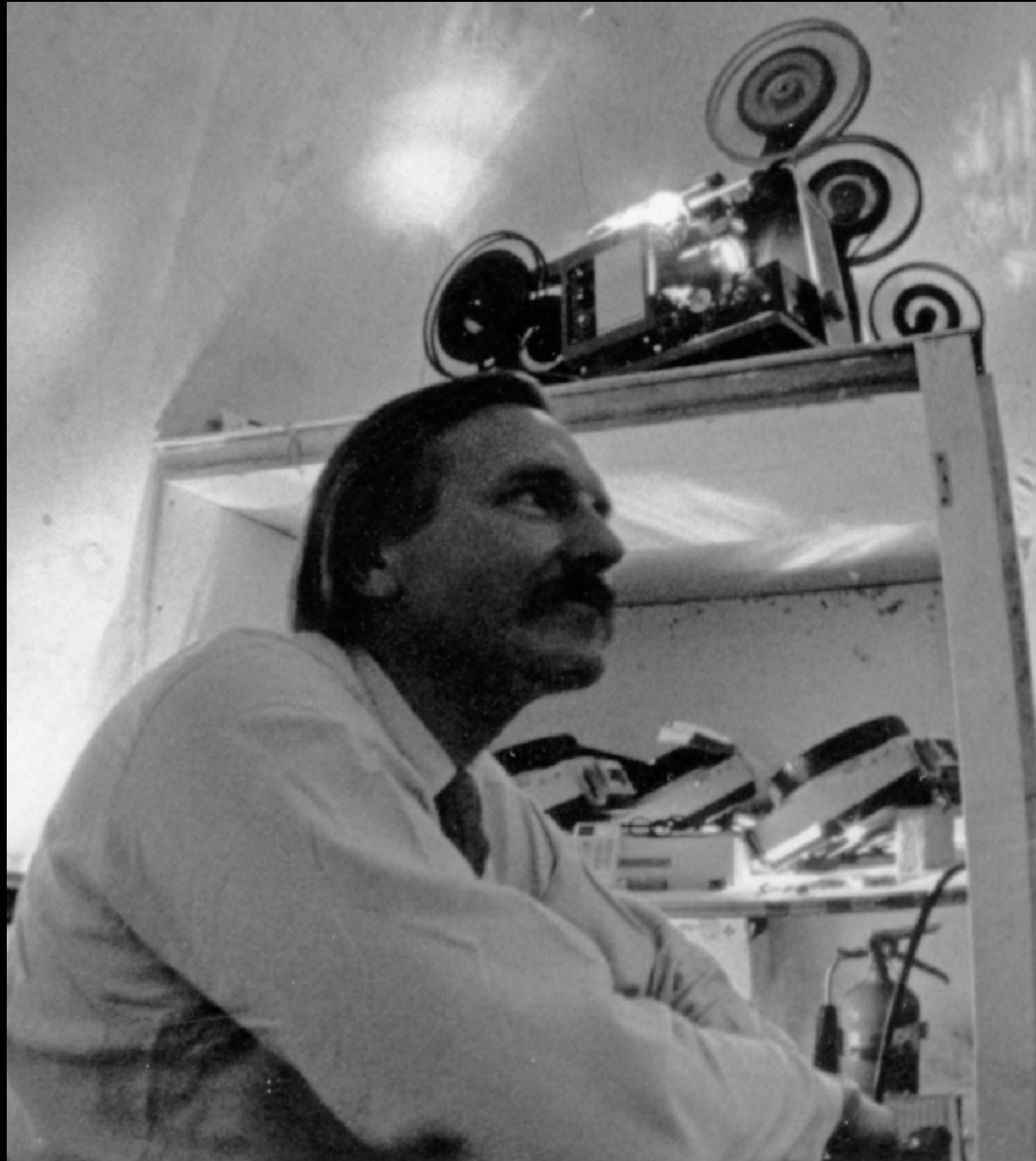


Gene Youngblood: Expanded Cinema,
New York: Dutton, 1970

“Echoing Fuller, Gene Youngblood, a loyal agent within the counterculture, identified the ‘electronic information network’ – what he referred to as the ‘videosphere’ or ‘global television’ – as the engine of this geopolitical transformation [i.e. end of nation states, N/A].”

Scott, Felicity D.: Architecture or Techno-Utopia. Politics after Modernism, Cambridge: MIT Press 2007, p. 202.

4 – Visual Music

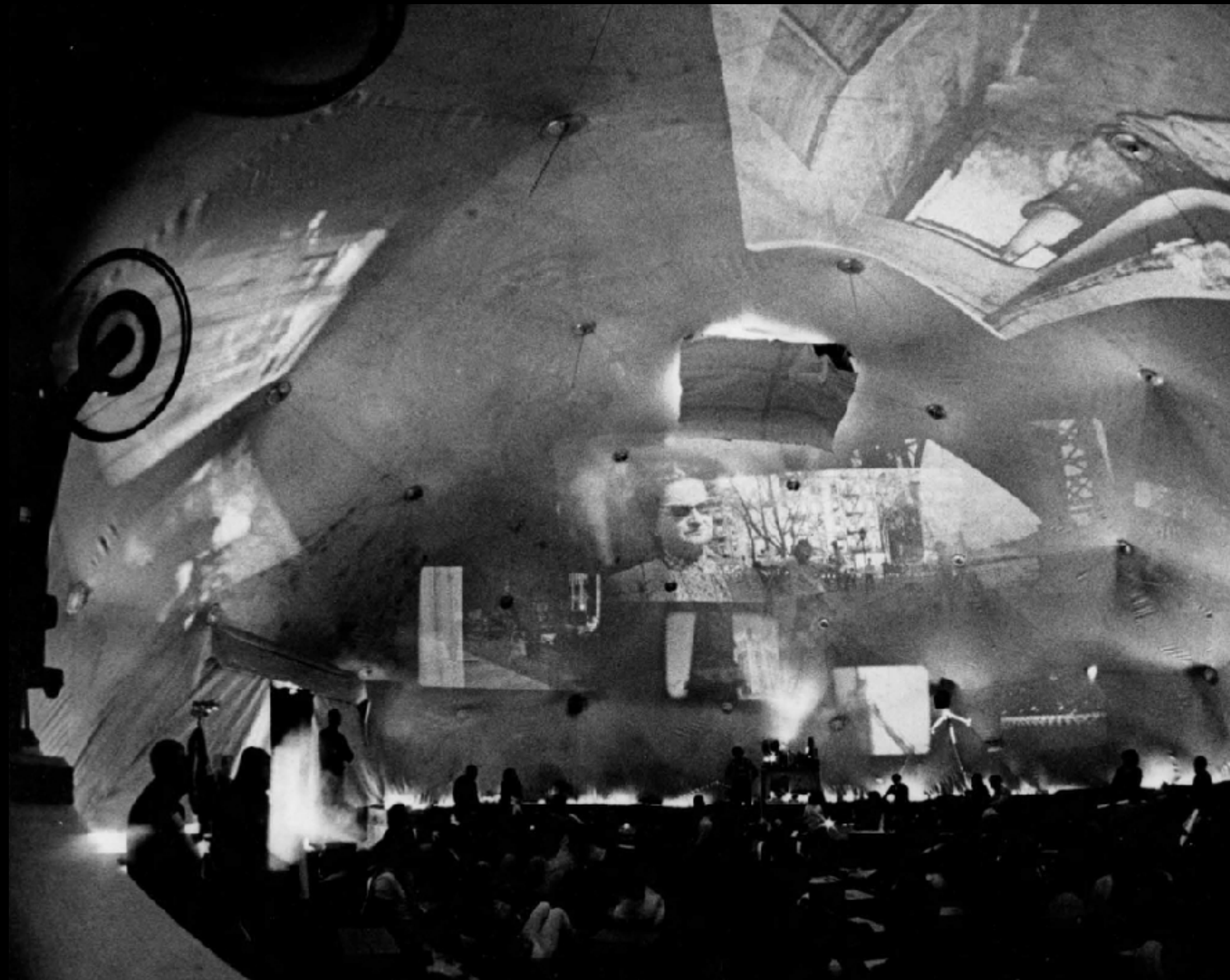


Stan VanDerBeek: Movie-Drome, 1963-1966
Stony Point, NY

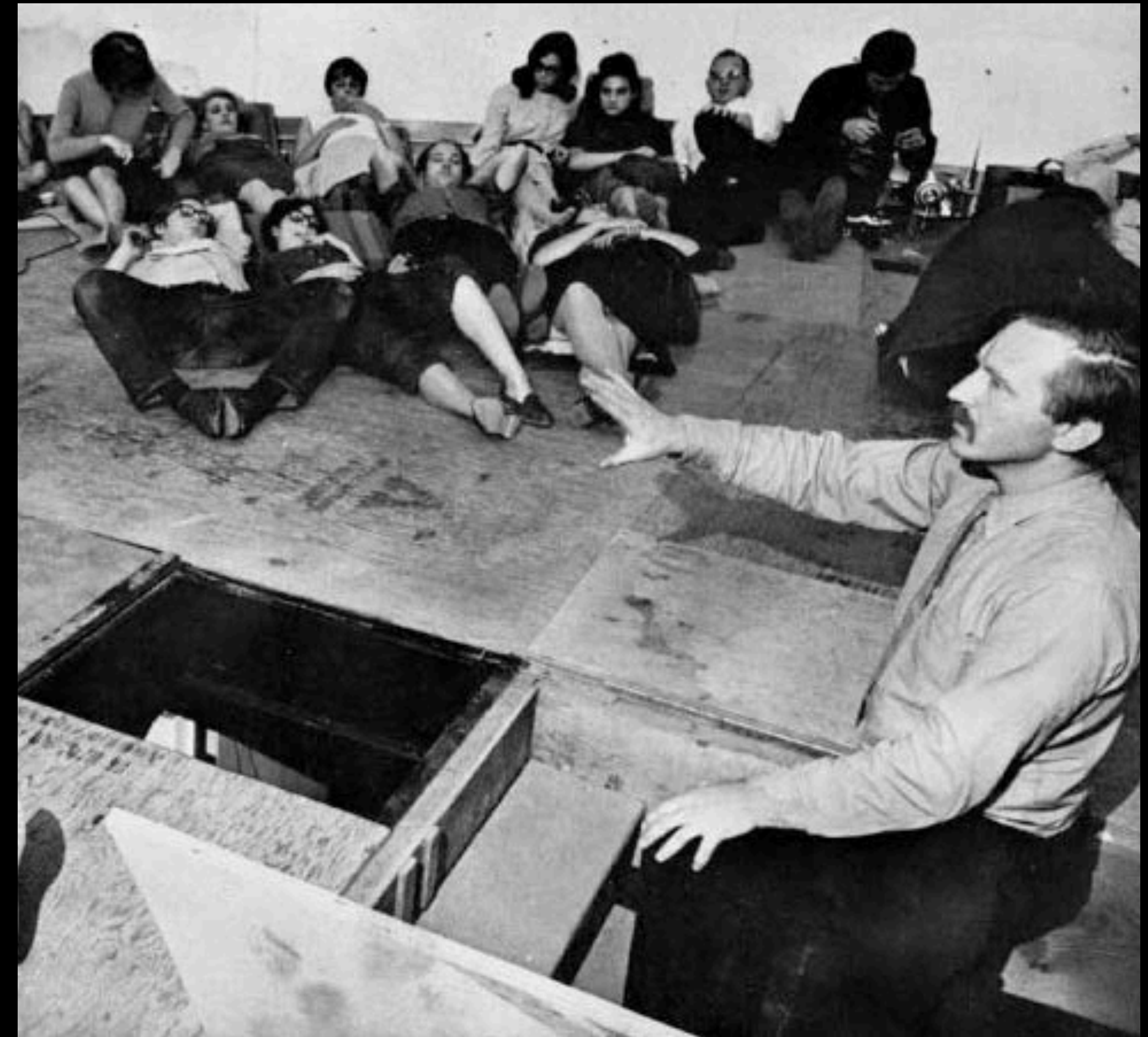
“non-verbal international language [...]. That immediate research begin on the possibility of a picture-language based on motion pictures. That we combine audio-visual devices into an educational tool: an experience machine or ‘culture-intercom.’”

VanDerBeek, Stan: „‘Culture: Intercom’ and Expanded Cinema: A Proposal and Manifesto“, *The Tulane Drama Review* 11/1 (1966), pp. 38–48, here pp. 39 and 41.

4 – Visual Music

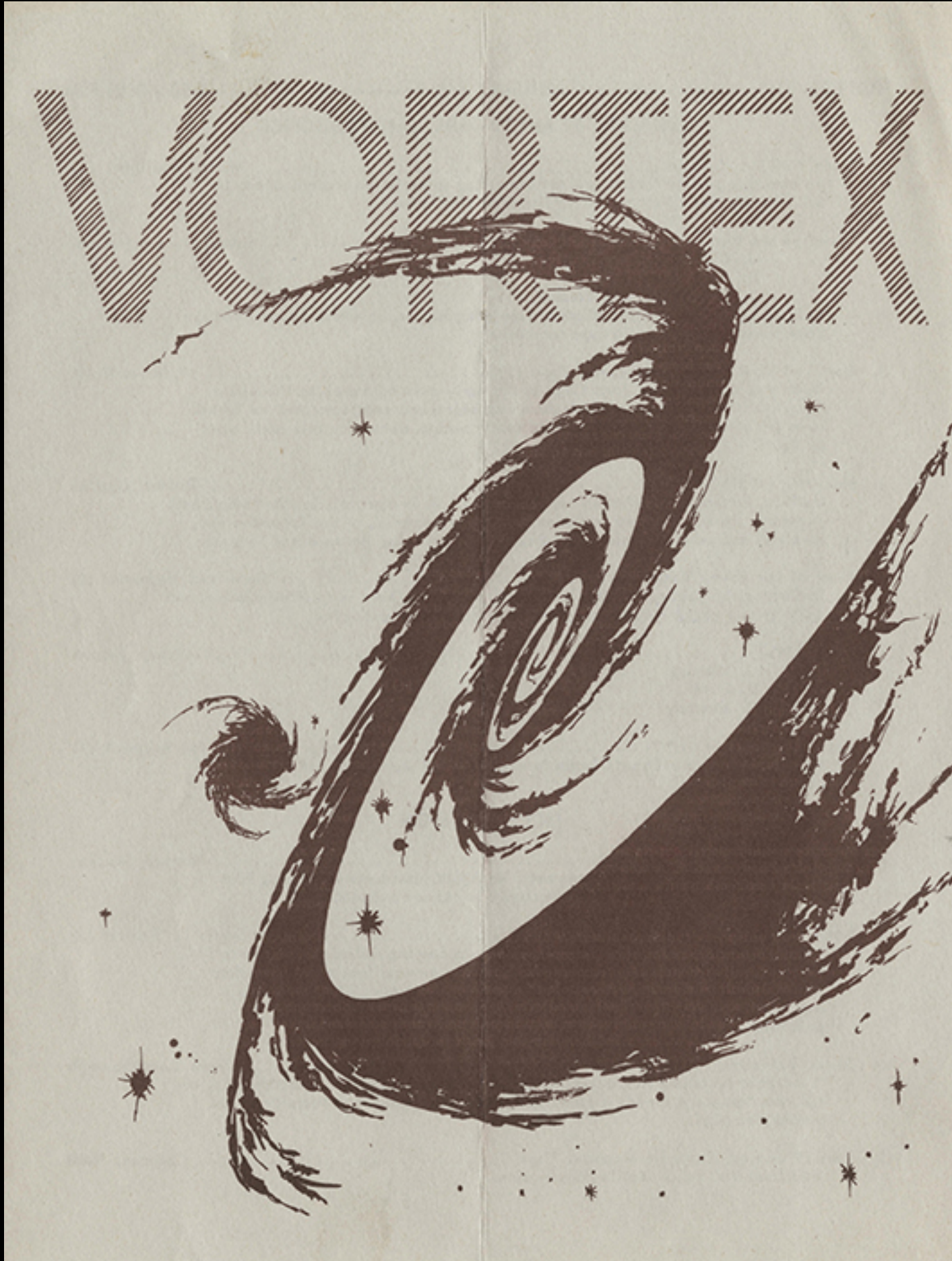


Stan VanDerBeek: Movie-Drome, 1963-1966

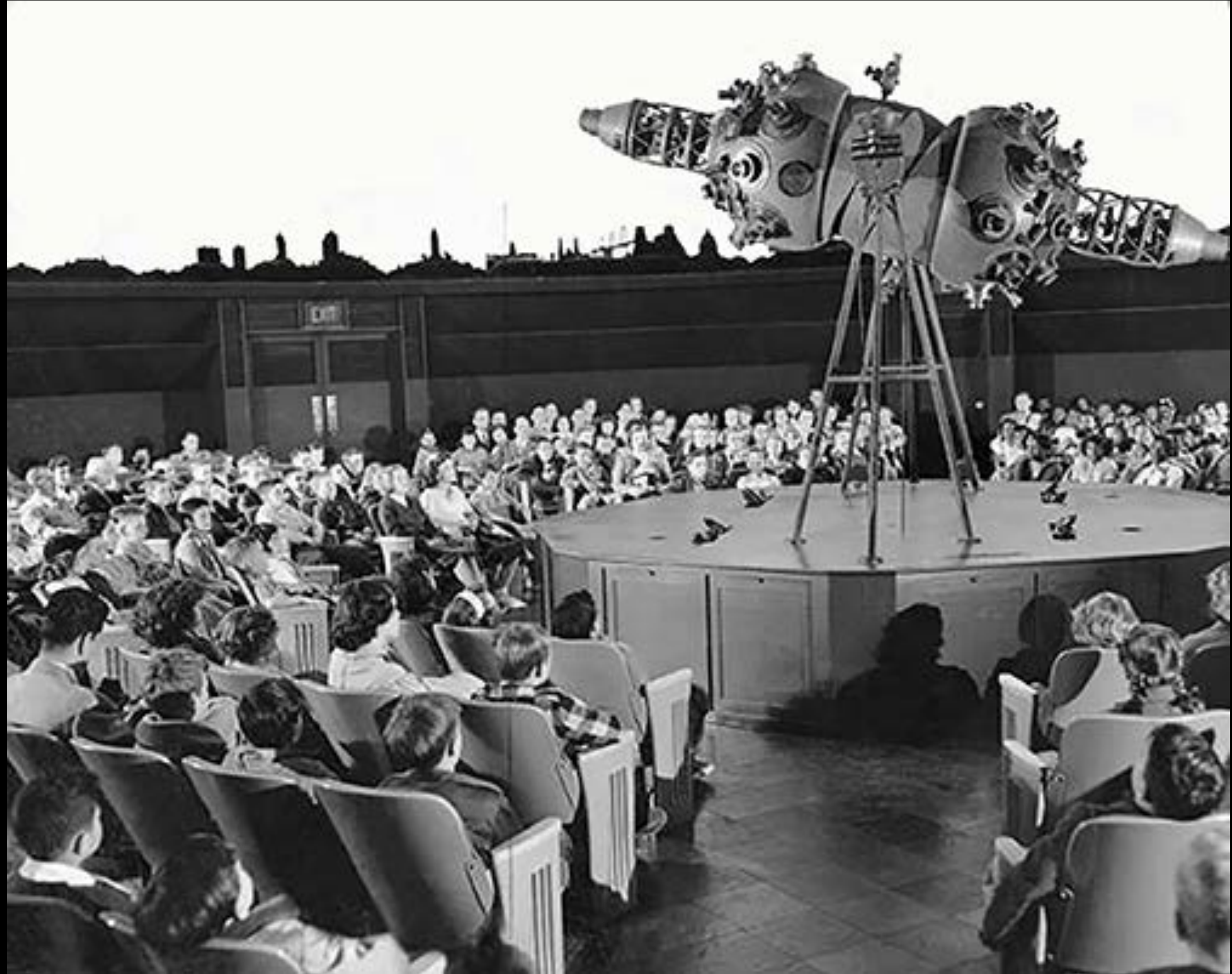


Stan VanDerBeek: Movie-Drome, 1963-1966
Stony Point, NY

4 – Visual Music

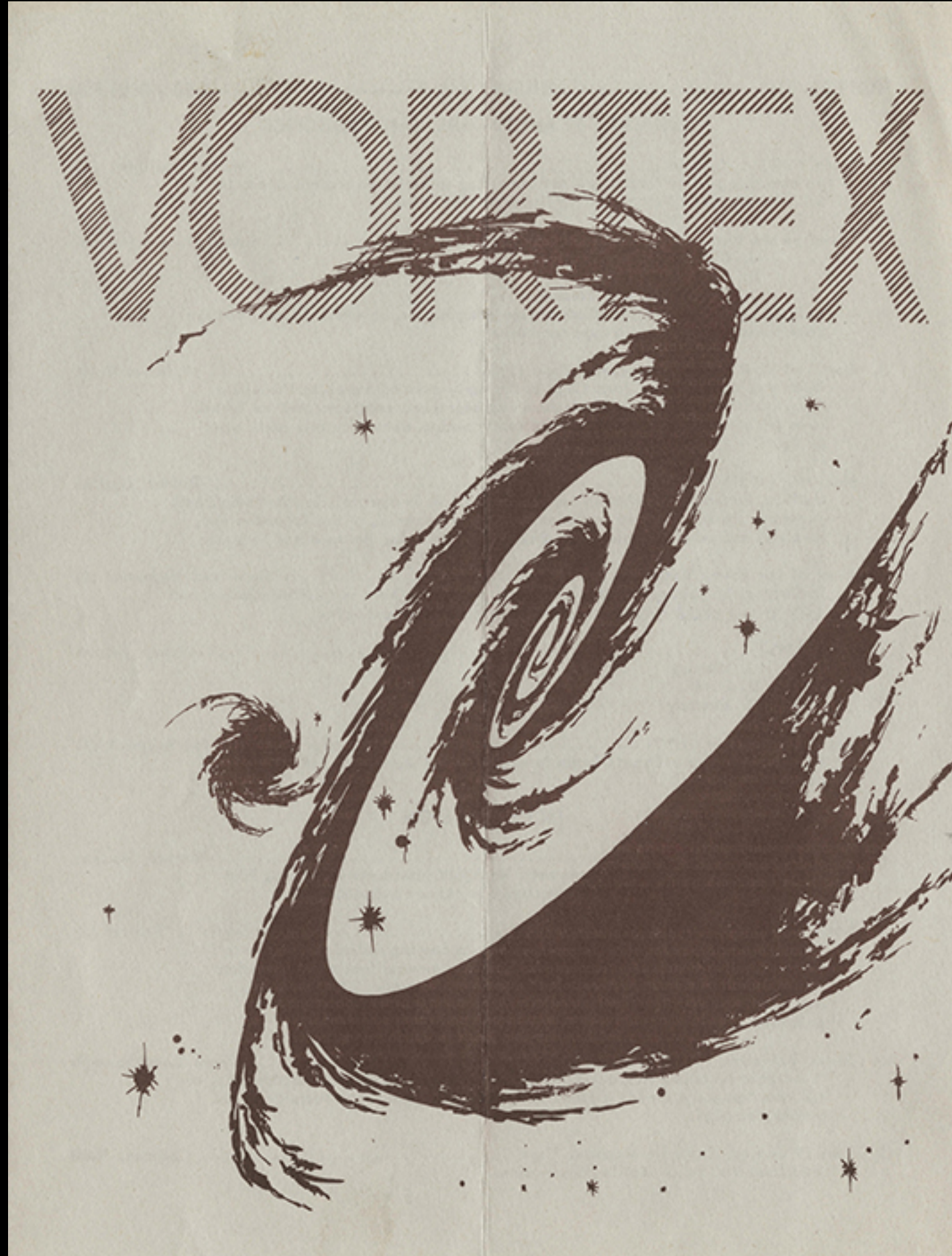


Vortex program flyer, 1958

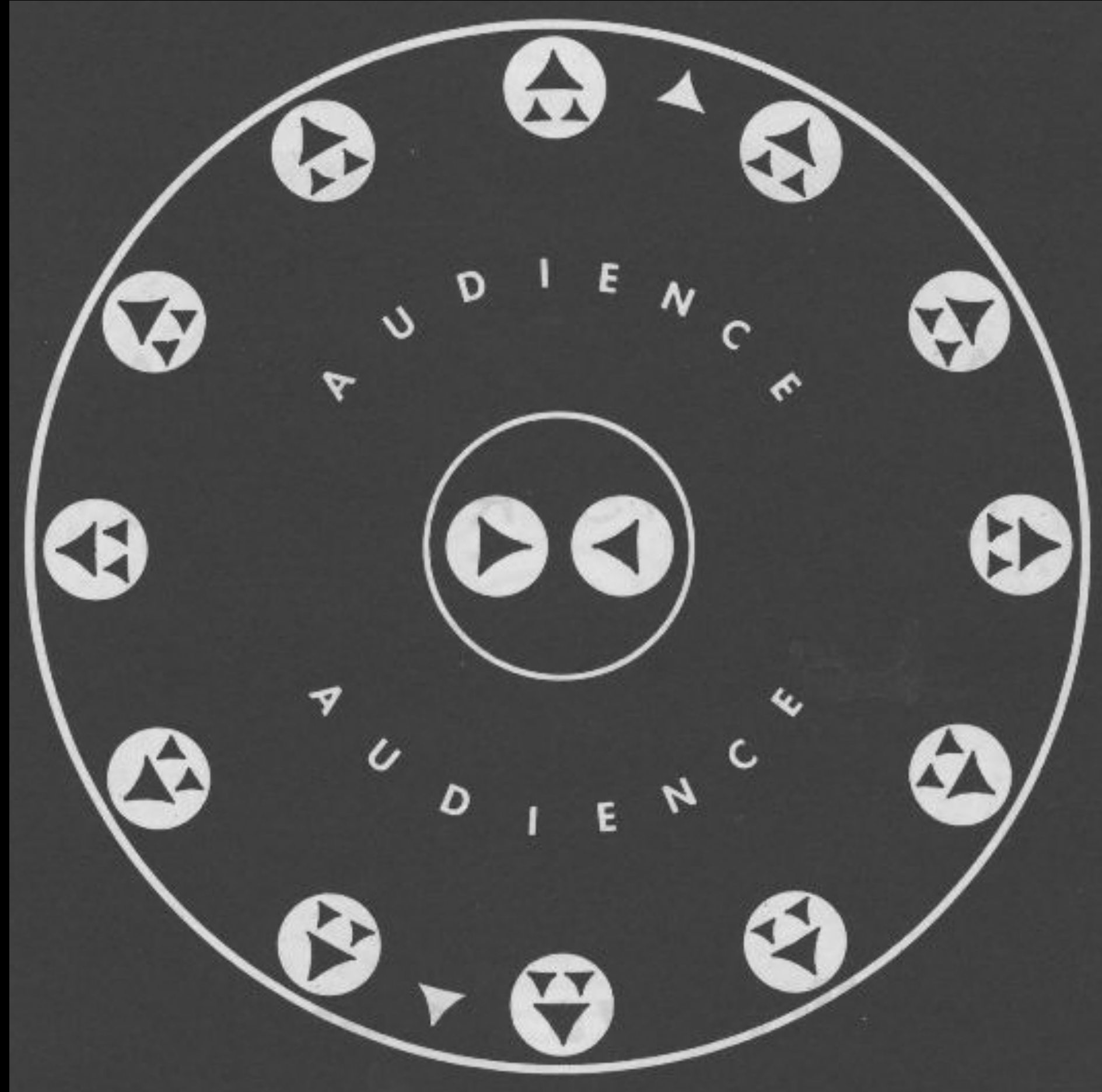


Morrison-Planetarium, San Francisco

4 – Visual Music



Vortex program flyer, 1958



Vortex liner notes, 1959

4 – Visual Music

VORTEX III----- JANUARY 6 & 7----- 8:00 - 9:15 P.M.

MORRISON PLANETARIUM, SAN FRANCISCO

1. Study (1953)..... Michel Philippot (1)
An exercise in pure form organized according to serial principles. *Study* is deliberately free of any poetic or literary association.

2. Notes on the History of a World. (1957)..... Gordon L. Longfellow
I. Birth and Perception
II. Evolution
III. Destruction and Torment
This is Longfellow's first effort at tape composition, in which he explored the combinations of piano and tape recorder.

3. Part I of Four Parts for Magnetic Tape (1953)..... Christian Wolff
Wolff was working with John Cage in the early days of American "musique concrete": Dancer Merce Cunningham choreographed and performed one of his most frightening, brilliant, shocking and humorous dances to this early work by Wolff.

4. Muskaku (1957)..... Robert Garfias
Garfias, a student of ethno-musicology at UCLA, is specializing in traditional Japanese music. *Muskaku* combines a multiple recording of the Japanese sho, hichiriki and an electronic drum. Engineering by Loughborough and Jacobs.

5. Four of the Seven Variations..... Moroi and Mayazumi (2)
Following the Japanese-influenced work of Garfias, here is a German-influenced work by two young electronic composer-engineers from Tokyo.

6. Time (1957)..... Henry Jacobs
I. Morning
II. Dusk
III. Evening

7. Rocking the Orbit (1957) Big Jay Sputnik (3)
Musique concrete elements begin to appear in the more profound works of the commercial school.

INTERMISSION

8. Think (1957)..... Norman Shacker
A synthetic composition for automatic keyboard instruments utilizing IBM scrolls for basic material. An experiment in chance composition.

9. Loop No. 3 (1957)..... David L. Talcott (4)
Loop is cut from half an hour of continuous recording utilizing the Ampex electronic feedback feature with two machines several feet apart, therefore increasing the delay to six seconds. As a sound source, Talcott used his own voice, rambling from idea to idea while the machines monitored the procedure.

10. Boom Revisited (1957).....William Loughborough
A multiple recording of Loughborough performing on percussion instruments of his own design. A steady rhythmic pattern is combined with variable speed percussive runs.

11. Part IV of Four Parts for Magnetic Tape Christian Wolff
The final section of Wolff's composition.

12. Composition for Piano and Tape Recorder (1954)..... Luening and Ussachevsky (5)
One of the more conservative tape compositions in which Columbia Professors Luening and Ussachevsky apply transposition (speed changes) and reverberation (echo effect) to a piano composition.

13. Excerpt from Sonata for Loudspeaker (1954)..... Henry Jacobs (6)
The Sonata evolved from the passing thought that rhythm has something to do with periodicity and repetition, and that a loudspeaker is generally more than permissive.

14. Aeolian Harp (the 20's)..... Henry Cowell (6)
Composer Cowell opened many a vista to new sounds with his approach to the conventional piano. In *Aeolian Harp*, he plucks the strings.

15. Jazz et Jazz (1954)..... Andre Hodeir
Hodeir, a French authority on American jazz, produced this rather amusing composition by combining rhythm and instrumentation in contrasting forward and reverse modes.

16. A,I (1956)..... Takemitsu (2)
A suggestive *tour de force* based on the Japanese word "love".

RECORDINGS USED

1. Ducretet-Thomson LP
No. 93121

2. Universal LP (Japanese)
No. PLP 1009

3. Imperial 45 rpm
No. X5480

4. MEA Stereotape No. 101

5. Phonotape PM-5007

6. Folkways LP
No. FX 6160

CREDITS

Visual Consultants Jordan Belson and George Buntin

Technical Consultants..... Alvin Gundred and Robert McCorkle

Publicity..... Gary Barrett

Posters George Furness and Mr. and Mrs. Raoul Guertin

Coordinator..... Henry Jacobs

Sponsors ... FM Station KPFA and the Morrison Planetarium of the California Academy of Sciences

A stereophonic tape has been produced by Musical Engineering Associates of highlights of Vortex I and II. It can be purchased for \$11.95 at the Planetarium or through Subscriber's Supply Service, KPFA.

Vortex III program, 1958

4 – Visual Music

VORTEX III----- JANUARY 6 & 7----- 8:00 - 9:15 P.M.

MORRISON PLANETARIUM, SAN FRANCISCO

1. Study (1953)..... Michel Philippot (1)
An exercise in pure form organized according to serial principles. *Study* is deliberately free of any poetic or literary association.

2. Notes on the History of a World. (1957)..... Gordon L. Longfellow
I. Birth and Perception
II. Evolution
III. Destruction and Torment
This is Longfellow's first effort at tape composition, in which he explored the combinations of piano and tape recorder.

3. Part I of Four Parts for Magnetic Tape (1953)..... Christian Wolff
Wolff was working with John Cage in the early days of American "musique concrete": Dancer Merce Cunningham choreographed and performed one of his most frightening, brilliant, shocking and humorous dances to this early work by Wolff.

4. Muskaku (1957)..... Robert Garfias
Garfias, a student of ethno-musicology at UCLA, is specializing in traditional Japanese music. *Muskaku* combines a multiple recording of the Japanese sho, hichiriki and an electronic drum. Engineering by Loughborough and Jacobs.

5. Four of the Seven Variations..... Moroi and Mayazumi (2)
Following the Japanese-influenced work of Garfias, here is a German-influenced work by two young electronic composer-engineers from Tokyo.

6. Time (1957)..... Henry Jacobs
I. Morning
II. Dusk
III. Evening

7. Rocking the Orbit (1957)..... Big Jay Sputnik (3)
Musique concrete elements begin to appear in the more profound works of the commercial school.

INTERMISSION

8. Think (1957)..... Norman Shacker
A synthetic composition for automatic keyboard instruments utilizing IBM scrolls for basic material. An experiment in chance composition.

9. Loop No. 3 (1957)..... David L. Talcott (4)
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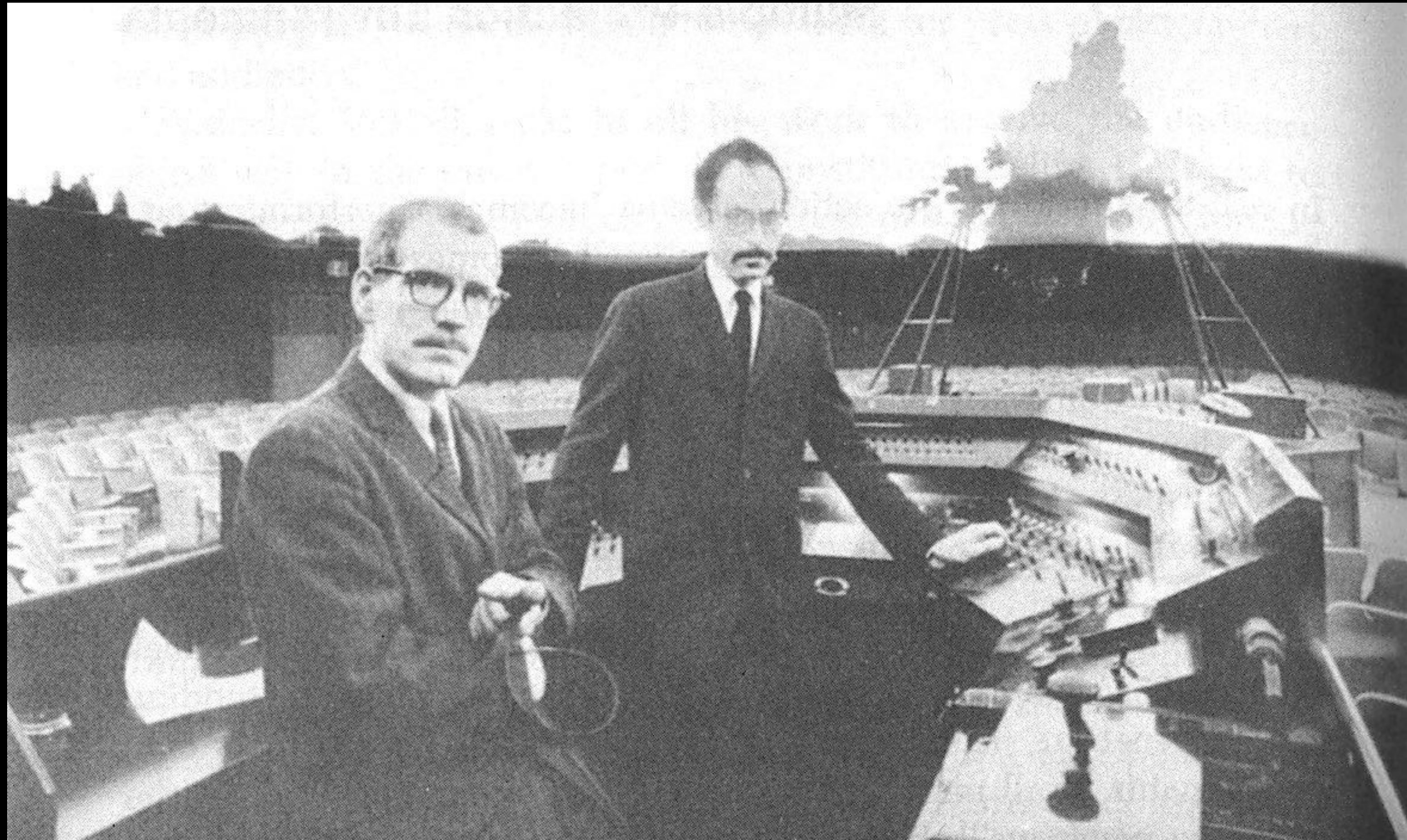
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Vortex III program, 1958

4 – Visual Music

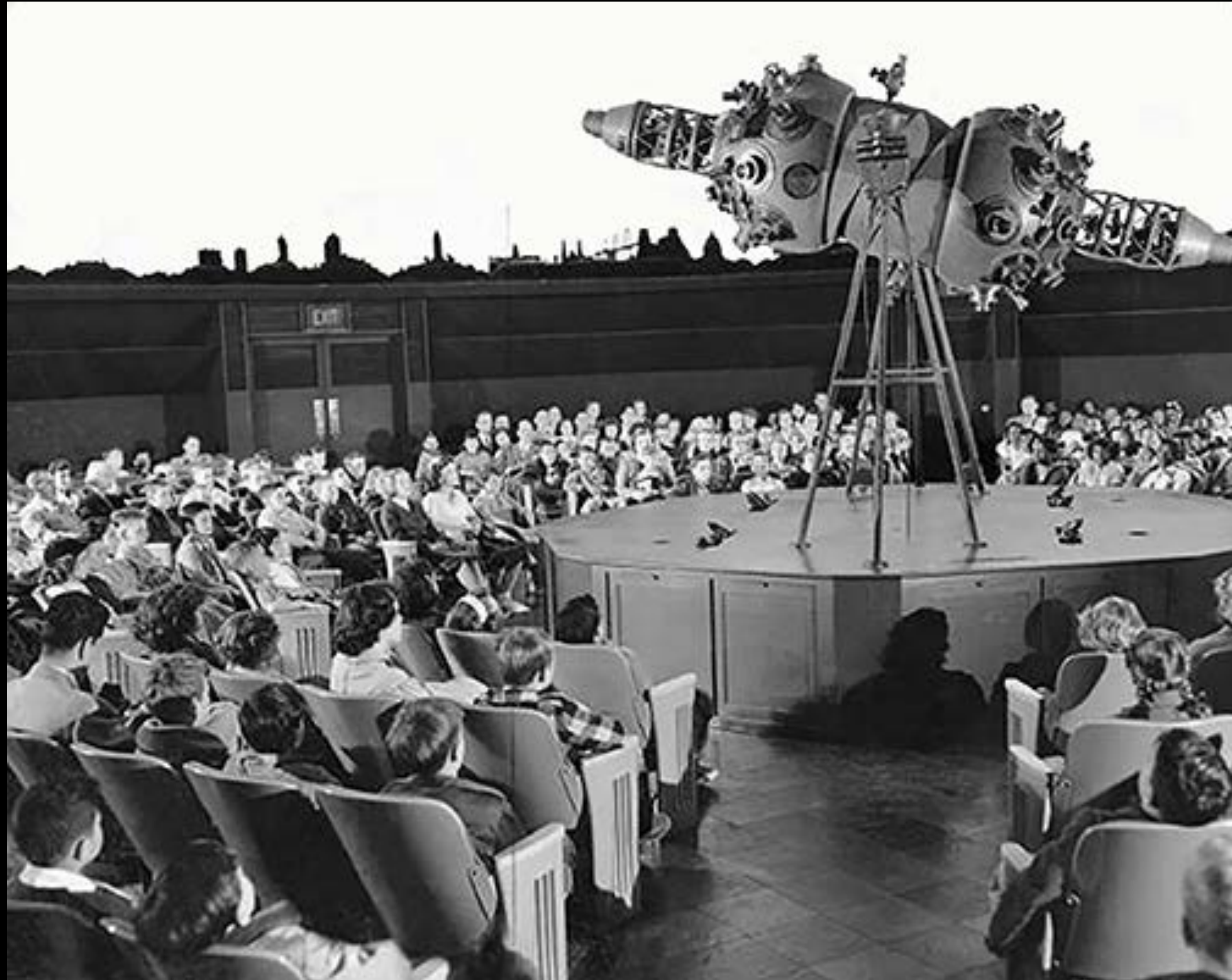


Henry Jacobs (left) and Jordan Belson at Morrison-Planetarium in San Francisco.

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Photo of star projector as published in:
Youngblood, Gene: Expanded Cinema,
New York: Dutton 1970, p. 390.



Morrison-Planetarium, San Francisco

4 – Visual Music

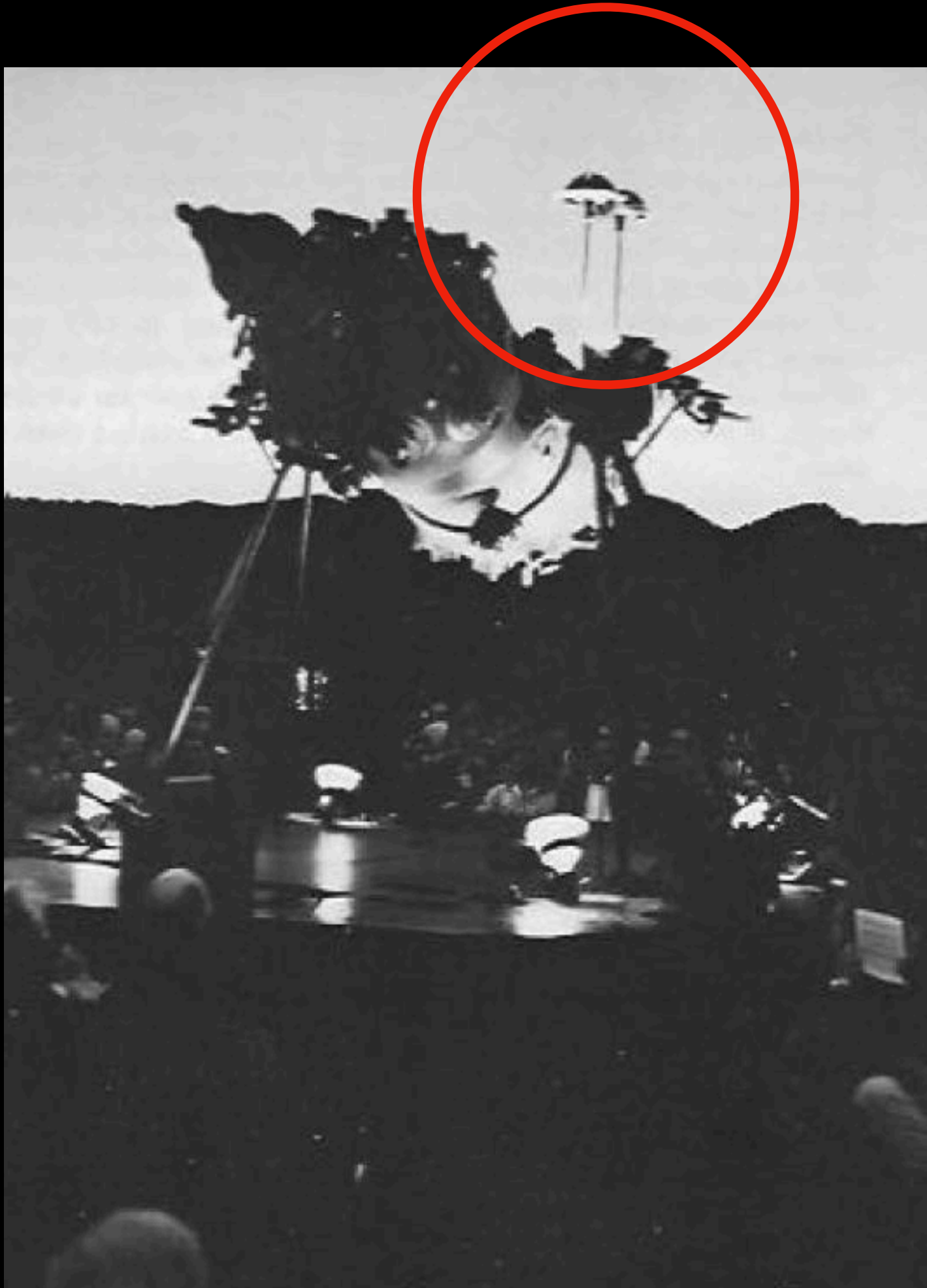
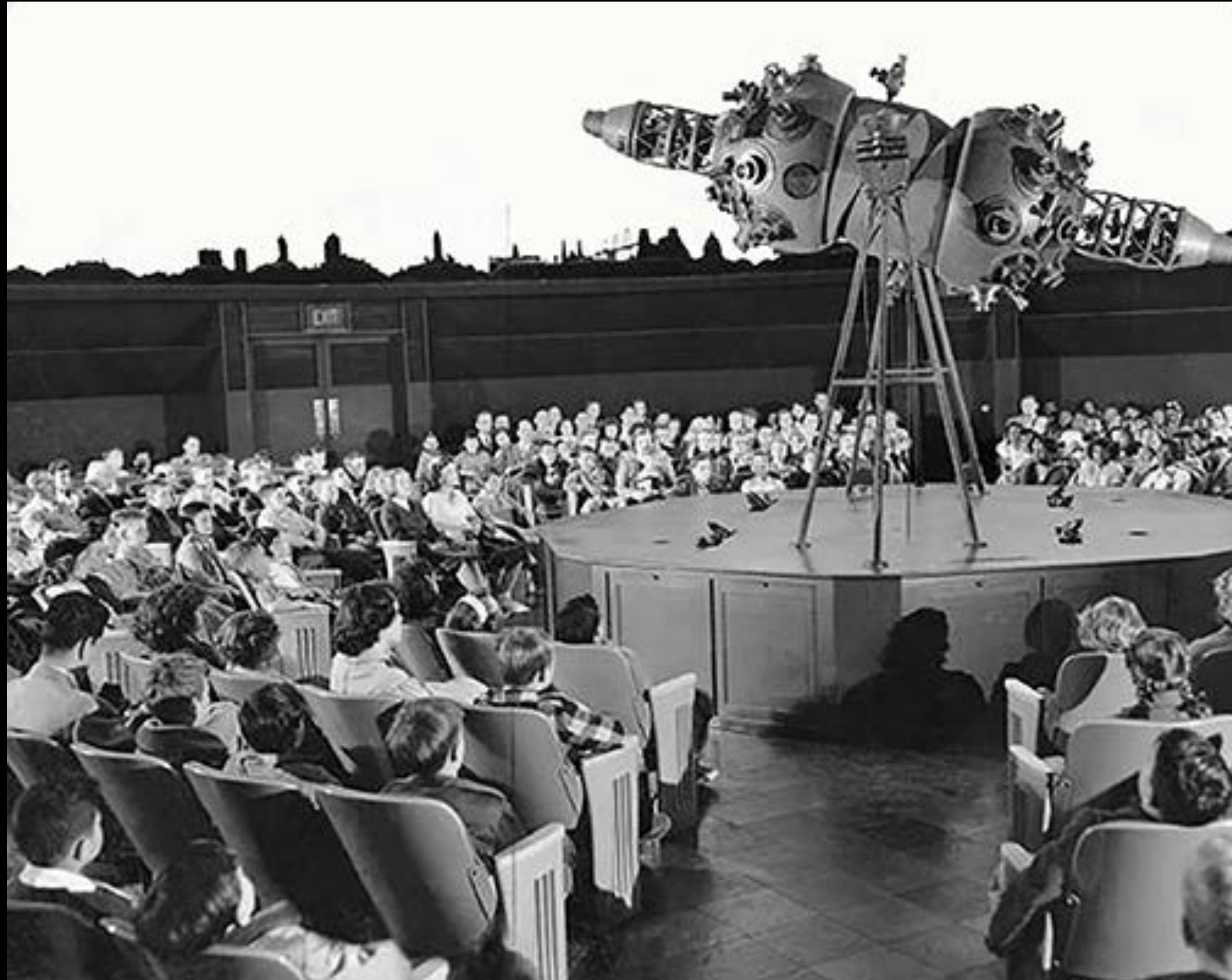
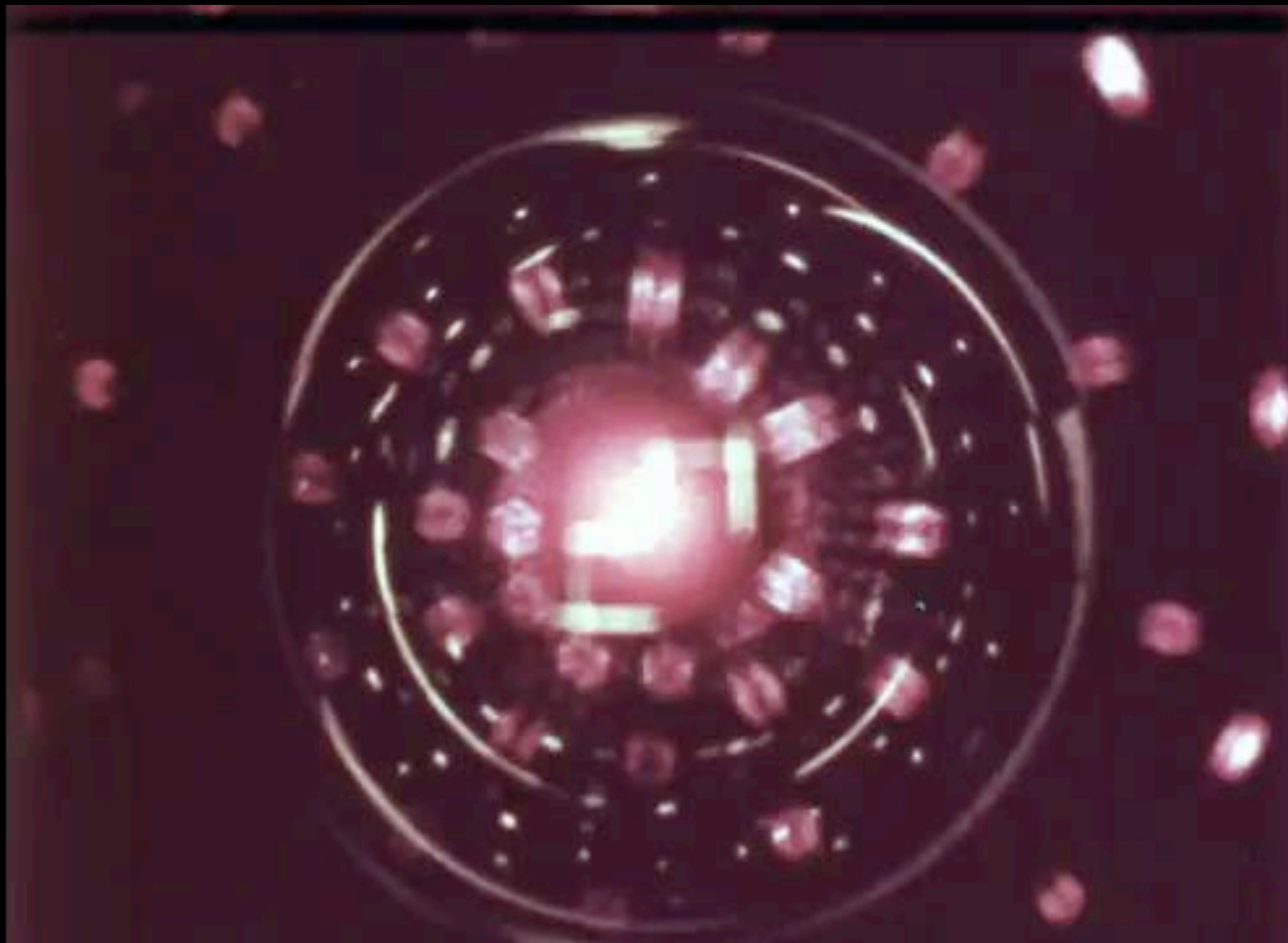
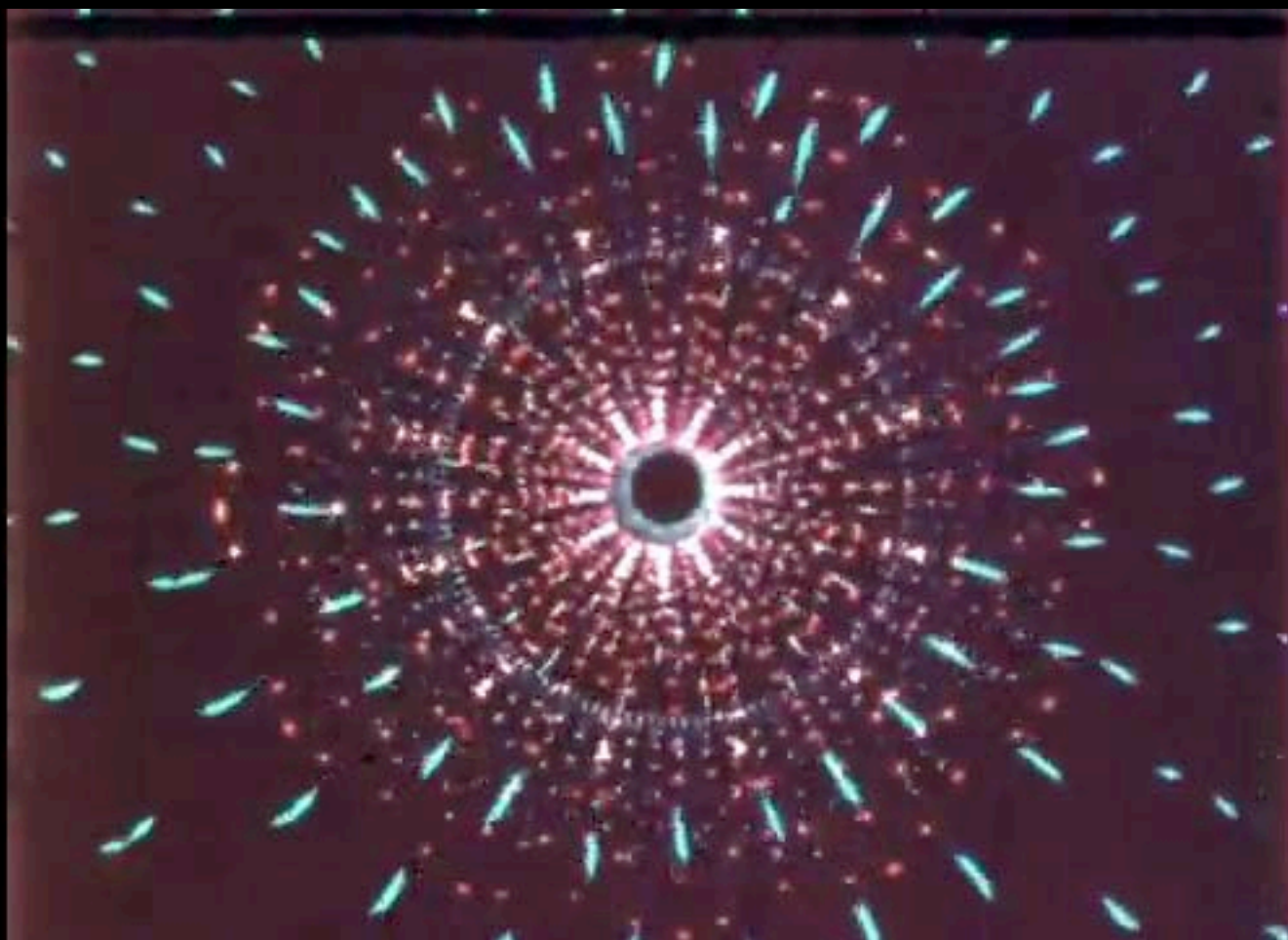
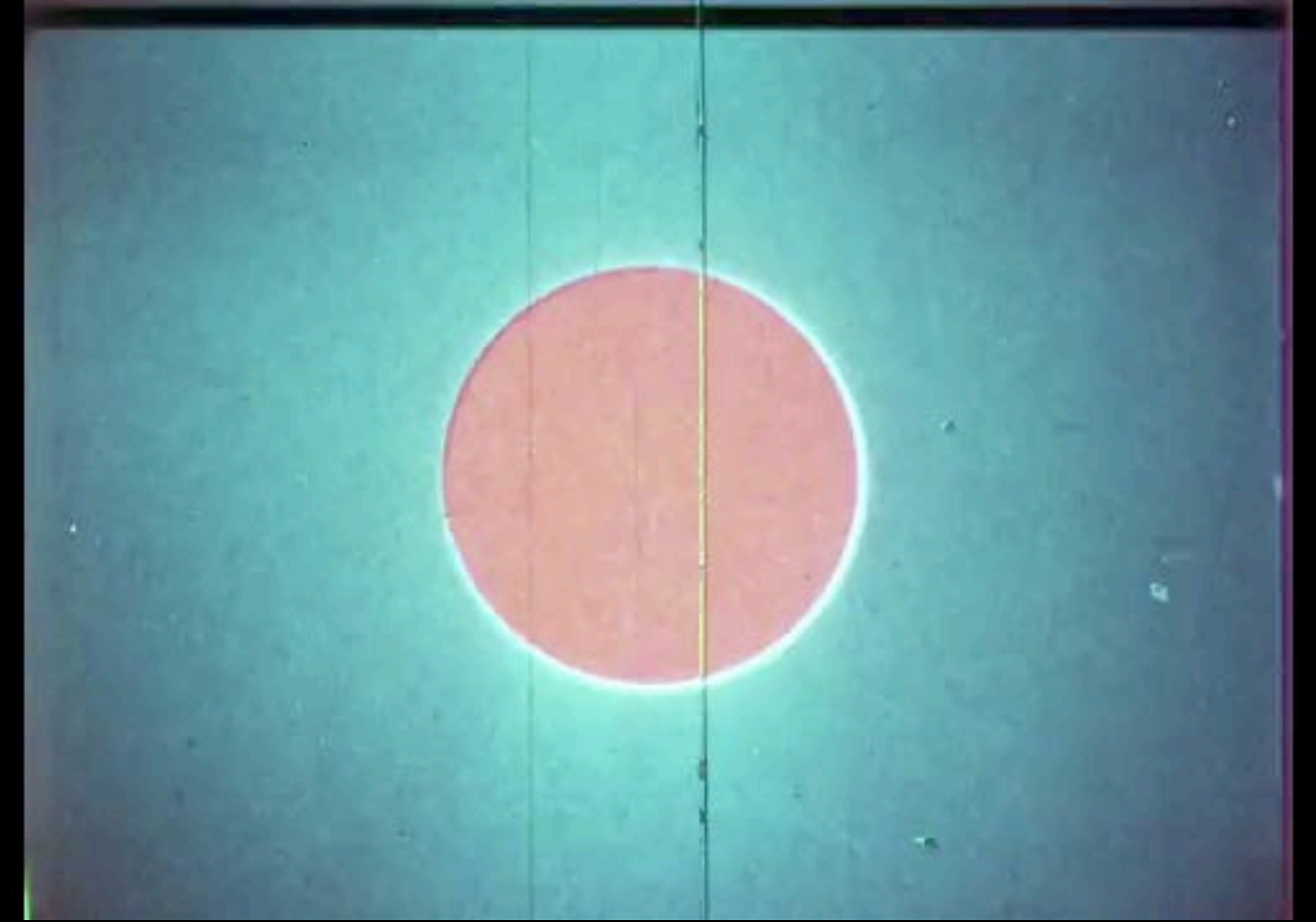
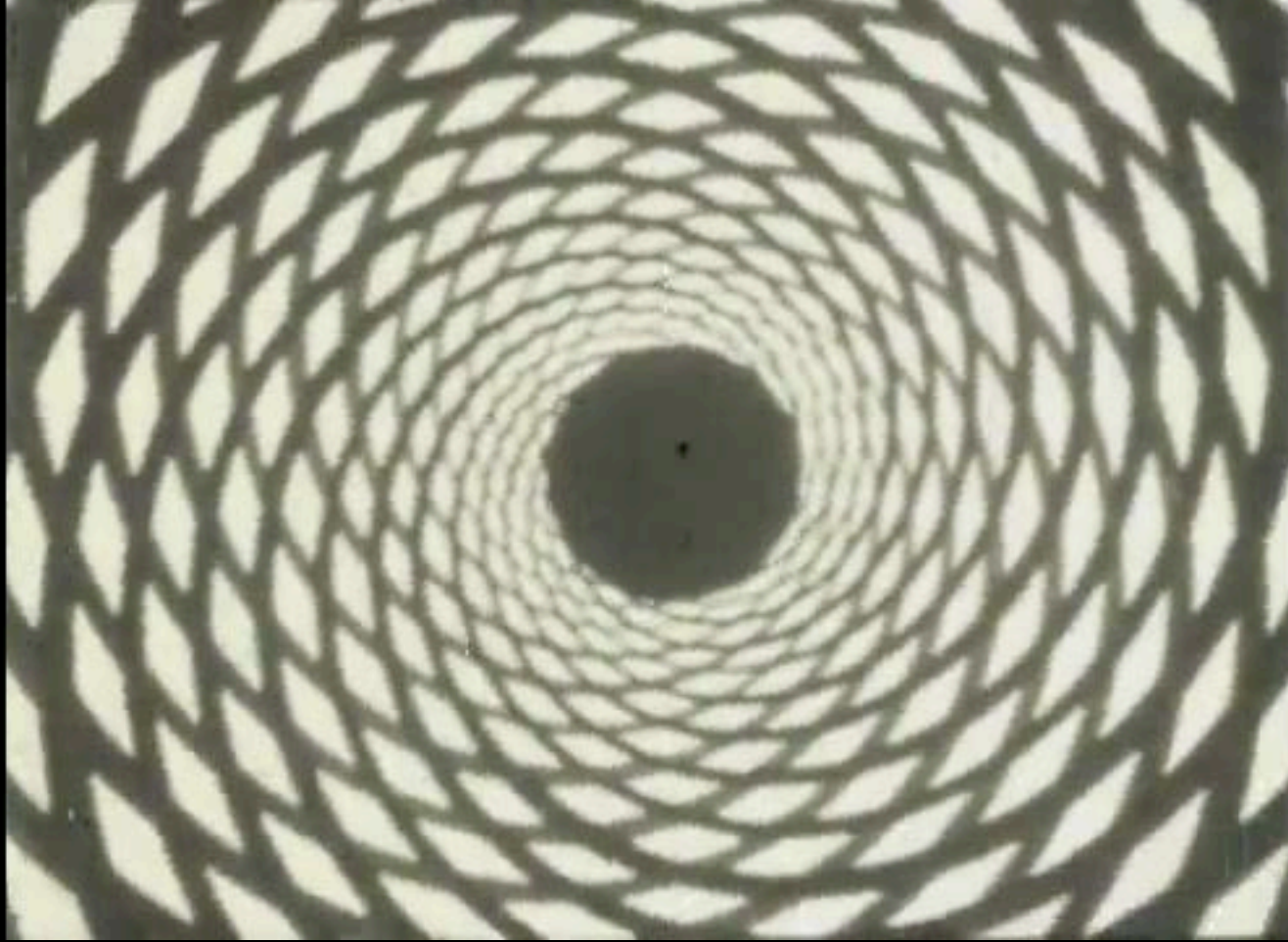
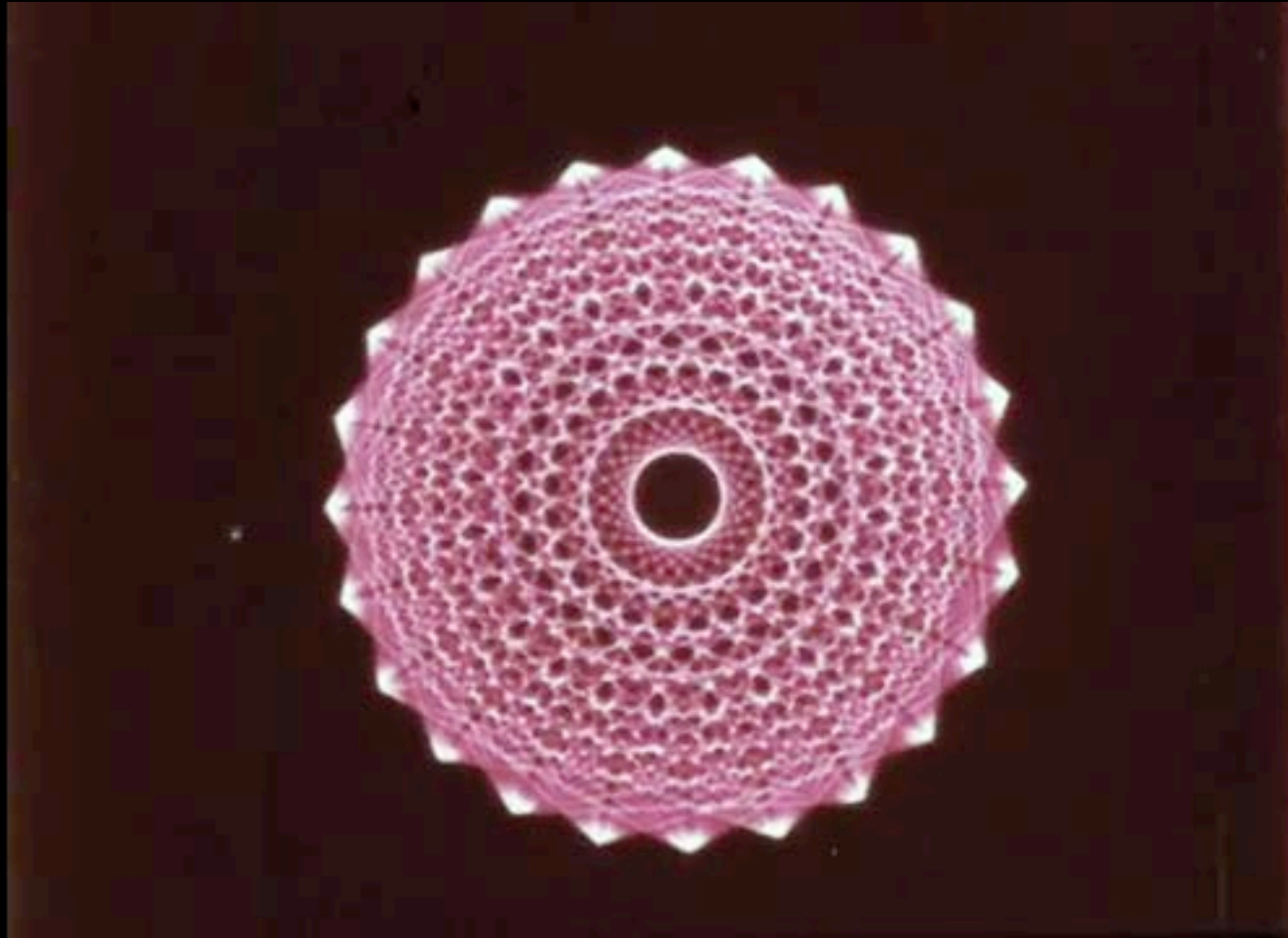


Photo of star projector as published in:
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New York: Dutton 1970, p. 390.

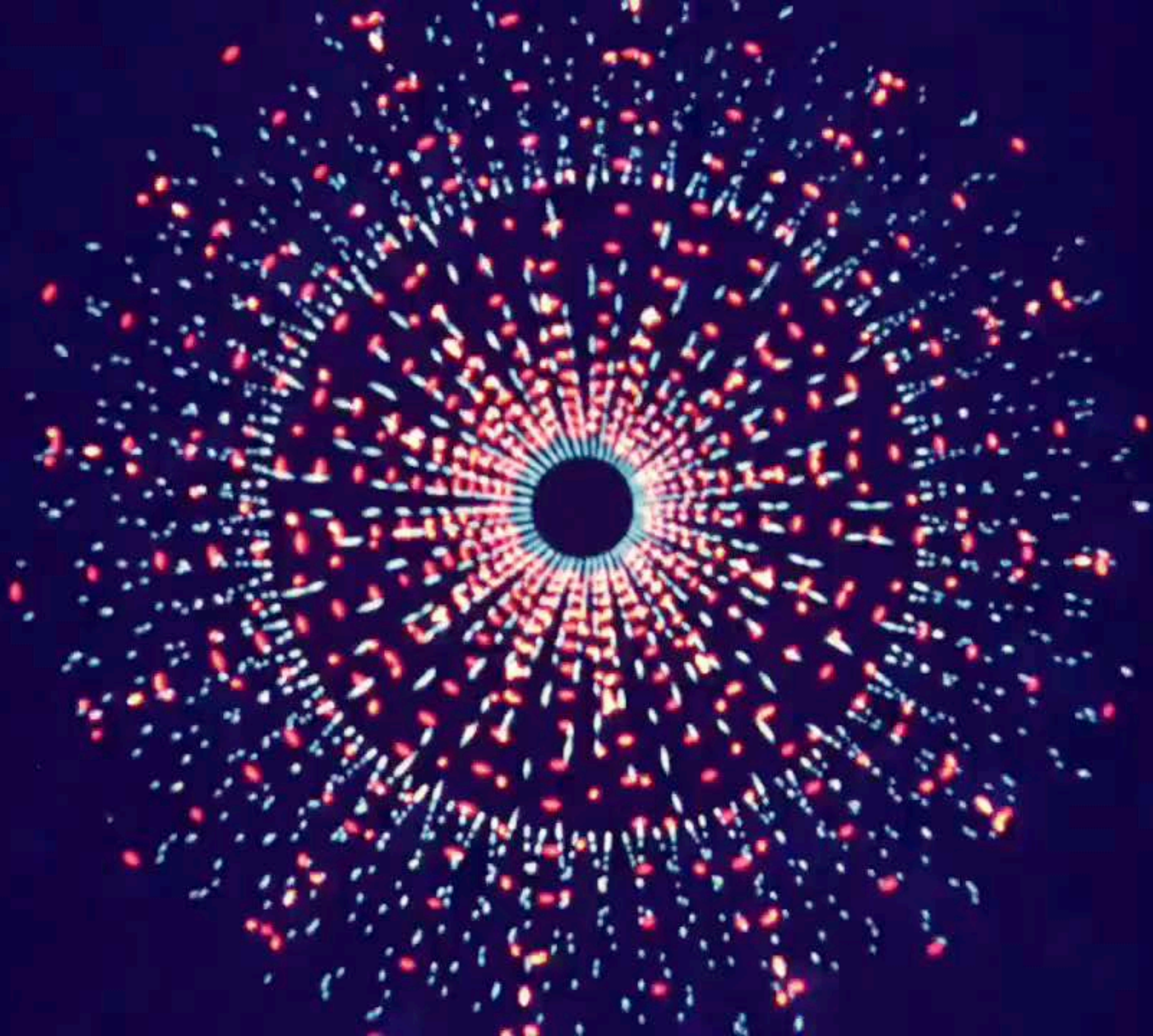


Morrison-Planetarium, San Francisco

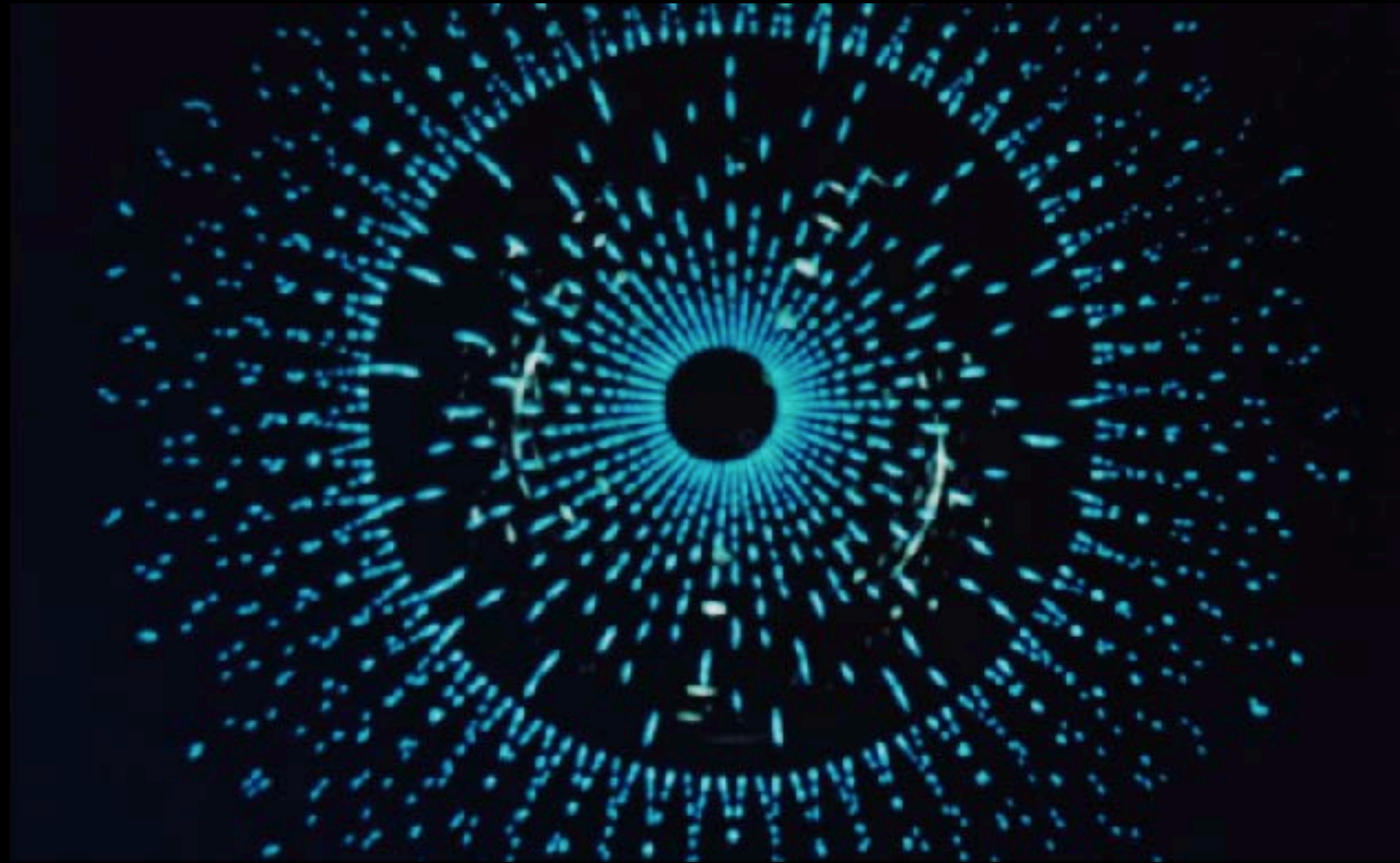
4 – Visual Music



Jordan Belson: Stills from Allures, 1961, 16mm, color, sound.



4 – Visual Music



Jordan Belson: Still from *Allures*, 1961
16mm, color, sound.

„Also we experimented with projecting images that had no motion-picture frame lines; we masked and filtered the light, and used images that didn't touch the frame lines. It had an uncanny effect: not only was the image free of the frame, but free of space somehow. It just hung there threedimensionally because there was no frame of reference.“

Interview with Jordan Belson, in: Youngblood, Gene: *Expanded Cinema*, New York: Dutton 1970, p. 389.

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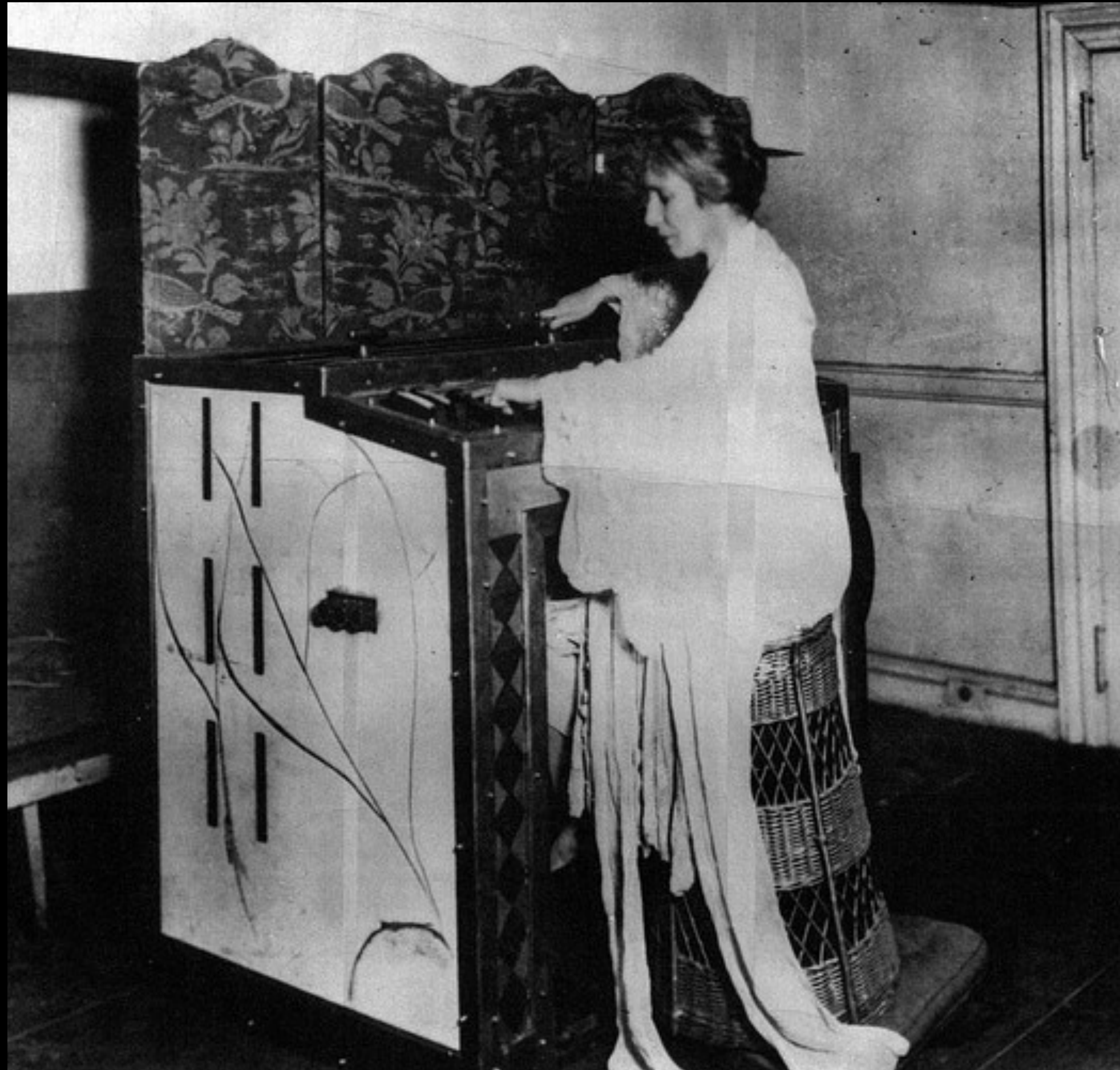


František Kupka: Amorpha, fugue en deux couleurs, 1912



Wassily Kandinsky: Impression III (Konzert), 1911

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Mary Hallock-Greenewalt, playing the Sarabet, 1925



Mary Hallock Greenewalt, at electric light "color organ", 1925

4 – Visual Music



Mary Hallock-Greenewalt, hand-painted transparency, ca. 1920, Historical Society of Pennsylvania.



Mary Hallock-Greenewalt, hand-painted scroll, ca. 1920, Historical Society of Pennsylvania.

4 – Visual Music

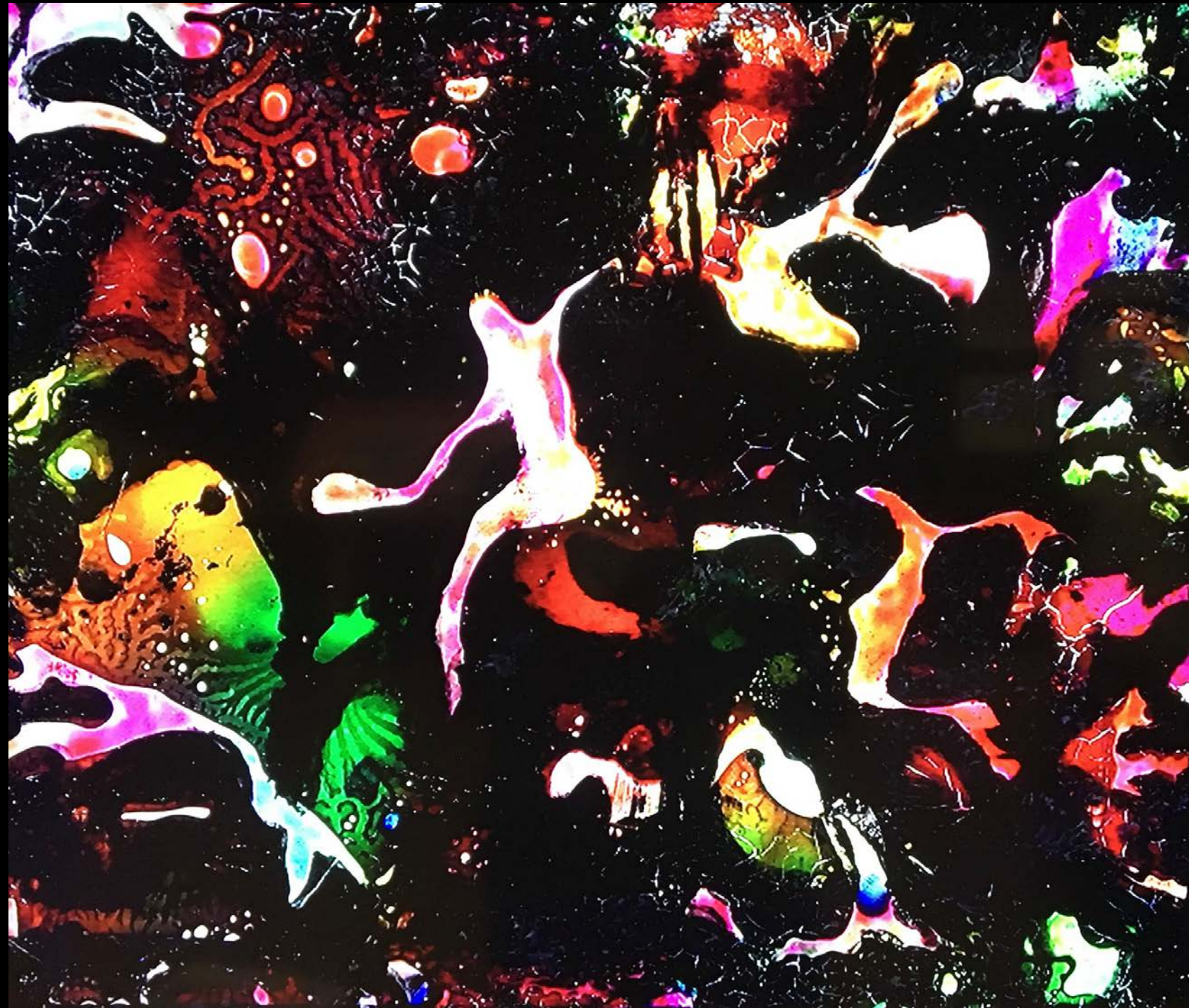


Mary Hallock-Greenewalt, playing the Sarabet, 1925

„Mary Hallock-Greenewalt became an electrical engineer in order to invent the Sarabet, her color organ. By the time of her death, she had received eleven patents for devices necessary for a variety of visual music she called Nourathar.“

Betancourt, Michael: „Mary Hallock-Greenewalt’s ‚Abstract Films‘“, *Millennium Film Journal* 45/46 (fall 2006), pp. 53–60, here p. 54

4 – Visual Music

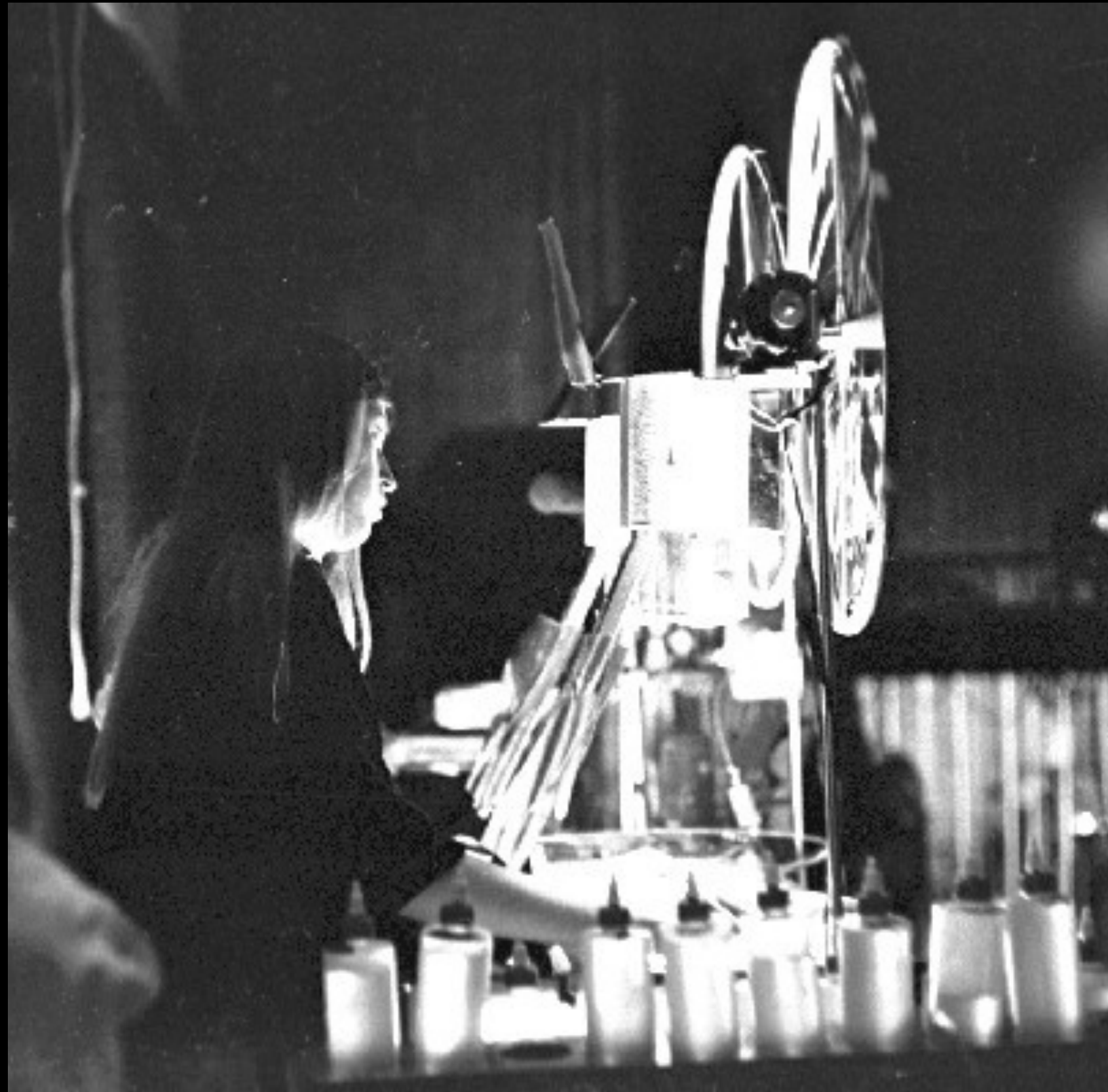


Sara Kathryn Arledge: glass slide transparency, ca. 1949, Berkeley Art Museum.

„These were projections that, unlike the Vortex shows, could be mobilized, and could use anything as their material support“.

Zinman, Gregory: Making Images Move. Handmade Cinema and the Other Arts, Berkeley: University of California Press 2020, p. 228

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Cecily Hoyt while mixing colored oils



Joshua Light Show (since 1968)

4 – Visual Music



„Künstlerische Entgrenzung, die Verbindung von Technologie und Ästhetik, kosmische Erweiterung des Bewusstseins, die Konvergenz der Sinne – alle diese Aspekte waren bereits in der Metapher der visuellen Musik angelegt. Der qualitative Unterschied, den sie in ihrer Umdeutung unter dem Eindruck von ‚cyberculture‘ und ‚expanded cinema‘ erfahren, ist der, dass ästhetische Effekte als physiologische Wirkungen definiert werden und die Auflösung des traditionellen Subjektbegriffs in eine soziale Utopie überführt wird.“

Engelke, Henning: Metaphern einer anderen Filmgeschichte.
Amerikanischer Experimentalfilm 1940–1960, Marburg: Schüren 2018, p. 162.